The time response of the system is the output of system as a func^n of time when subjected to a known input.
The time response of a control system is usually divided into two parts :-

* Transient Response
* Steady State response

$$\boxed{c(t) = C_{tr}(t) + C_{ss}(t)}$$

where, $c(t)$ = total time response
$C_{tr}(t)$ = transient time-response
$C_{ss}(t)$ = steady-state response.

TRANSIENT RESPONSE:

It is defined as the part of the time response that goes to zero as time becomes very large.

$$\lim_{t \to \infty} C_{tr}(t) = 0$$

It is also known as dynamic response of the system.

STEADY-STATE RESPONSE

The steady-state response of the system is the response of the system for a given input that remains after the transients have died out.

STANDARD TEST INPUTS

1.) Step function:-

The step func^n is given by

$$r(t) = \begin{cases} 0; & t<0 \\ A; & t \geq 0 \end{cases}$$

If $A=1$
$$r(t) = u(t)$$

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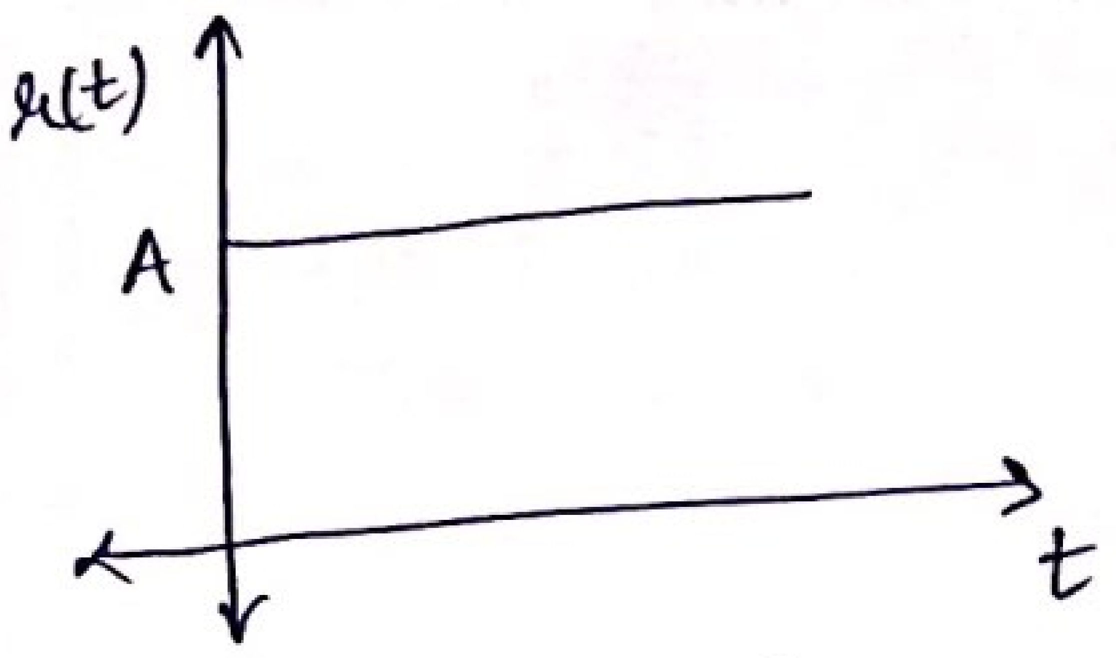
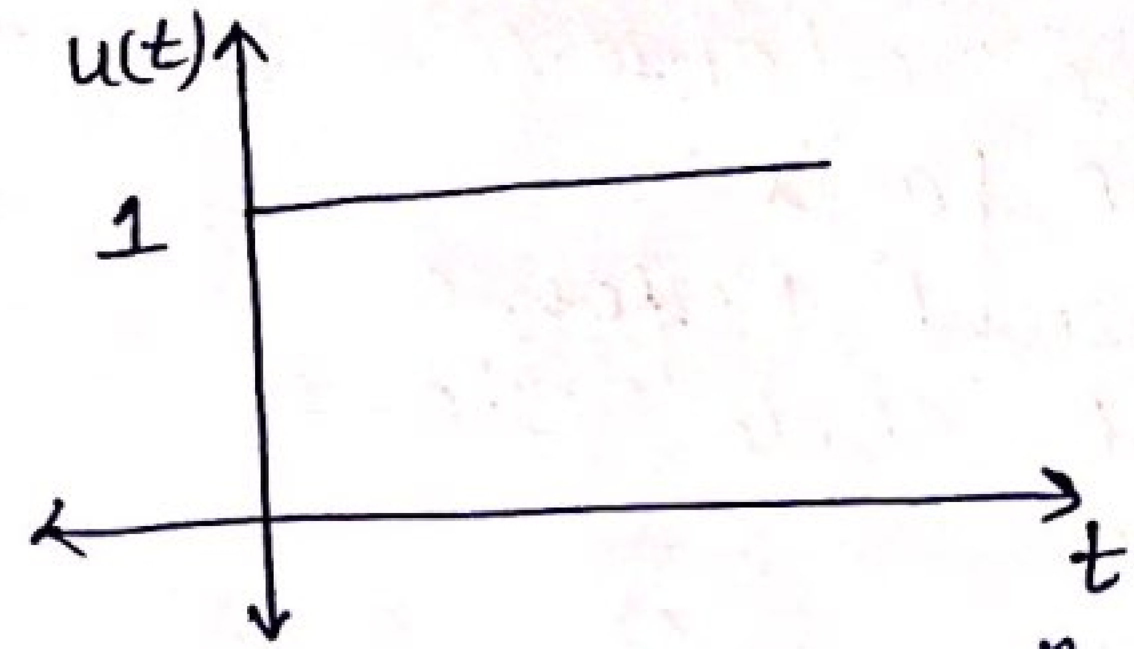
$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

It is known as unit step funcⁿ.



Step funcⁿ



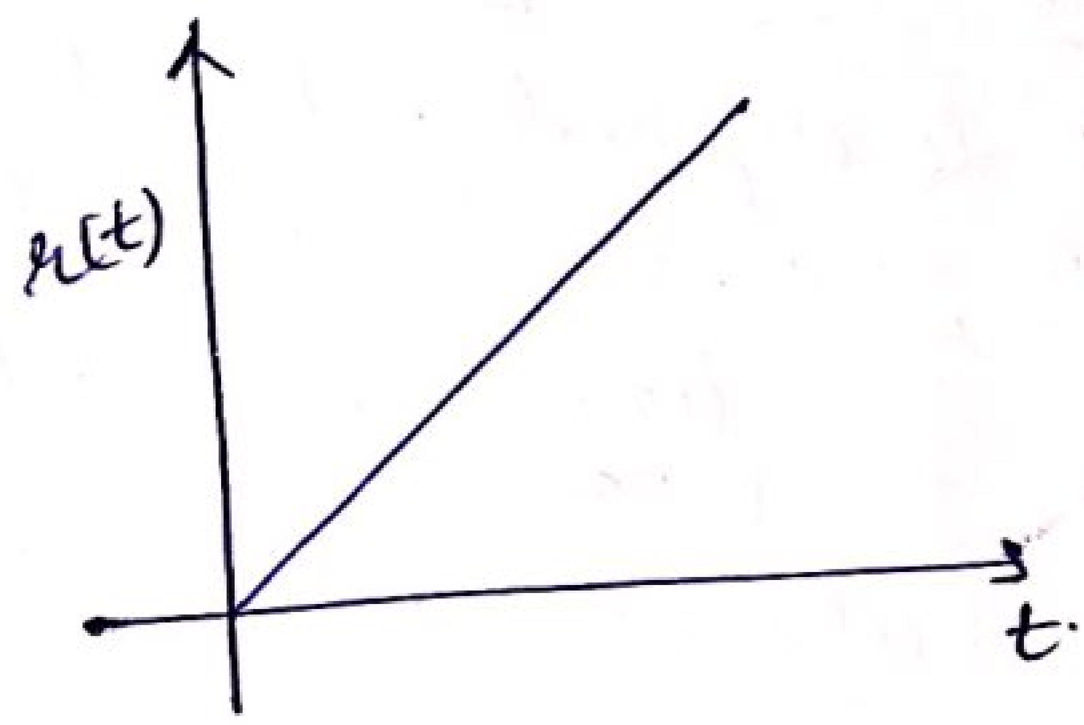
unit step funcⁿ

* The laplace transform of unit step funcⁿ is

$$R(s) = \frac{1}{s}.$$

2) Ramp funcⁿ :-

$$r(t) = \begin{cases} 0, & t \leq 0 \\ kt, & t > 0 \end{cases}$$



where k is the slope of the line

If $K = 1$,

$$r(t) = t$$

It is known as the unit ramp funcⁿ

* The laplace transform of unit step funcⁿ is given by:-

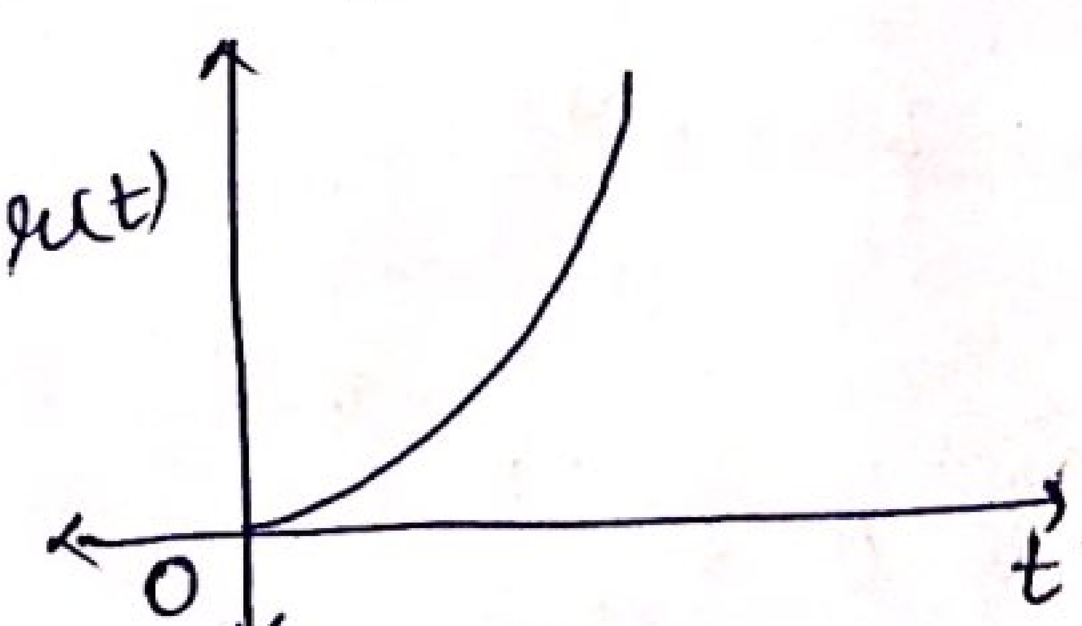
$$R(s) = \frac{1}{s^2}$$

* Ramp funcⁿ is also known as Velocity funcⁿ

3) Parabolic funcⁿ :-

The unit parabolic funcⁿ is given by :-

$$r(t) = \begin{cases} 0, & t \leq 0 \\ \frac{t^2}{2}, & t > 0 \end{cases}$$



* The Laplace transform of

$$R(s) = \frac{1}{s^3}.$$

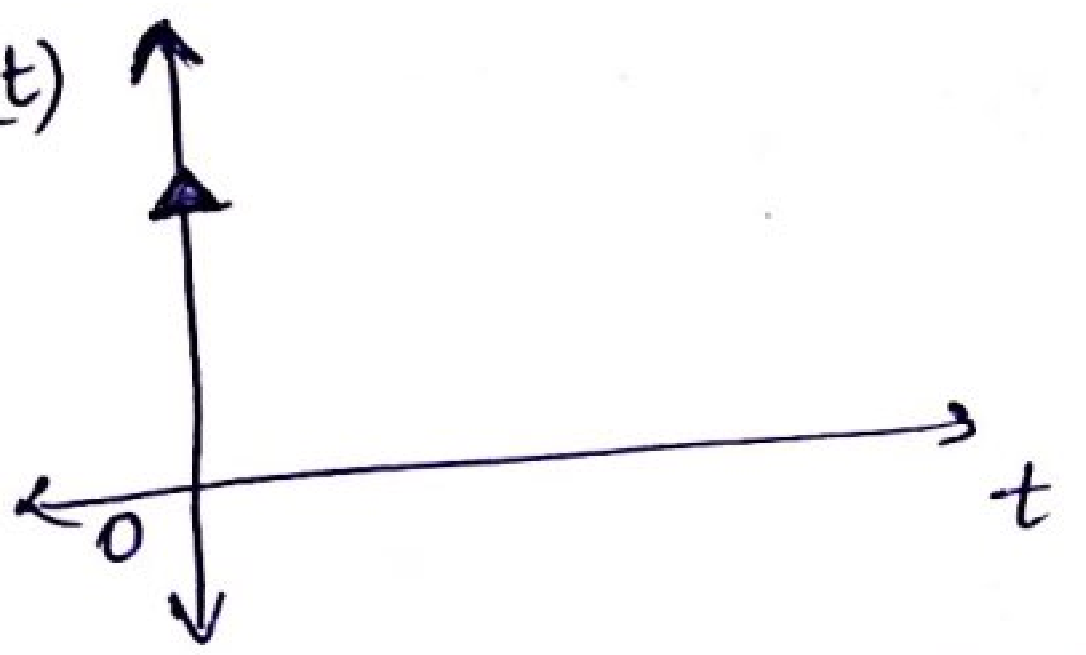
* The parabolic func^n is also called the acceleration func^n.

4.7 Impulse Func^n:-

Unit Impulse func^n is defined as :-

$$\left.\begin{aligned} \delta(t) &= 0, \quad t \neq 0 \\ \delta(t) &= \infty, \quad t = 0 \end{aligned}\right\}$$

* Area under the impulse is $\int_{-\infty}^{\infty} \delta(t)\,dt = 1$.



TIME RESPONSE OF FIRST ORDER SYSTEM

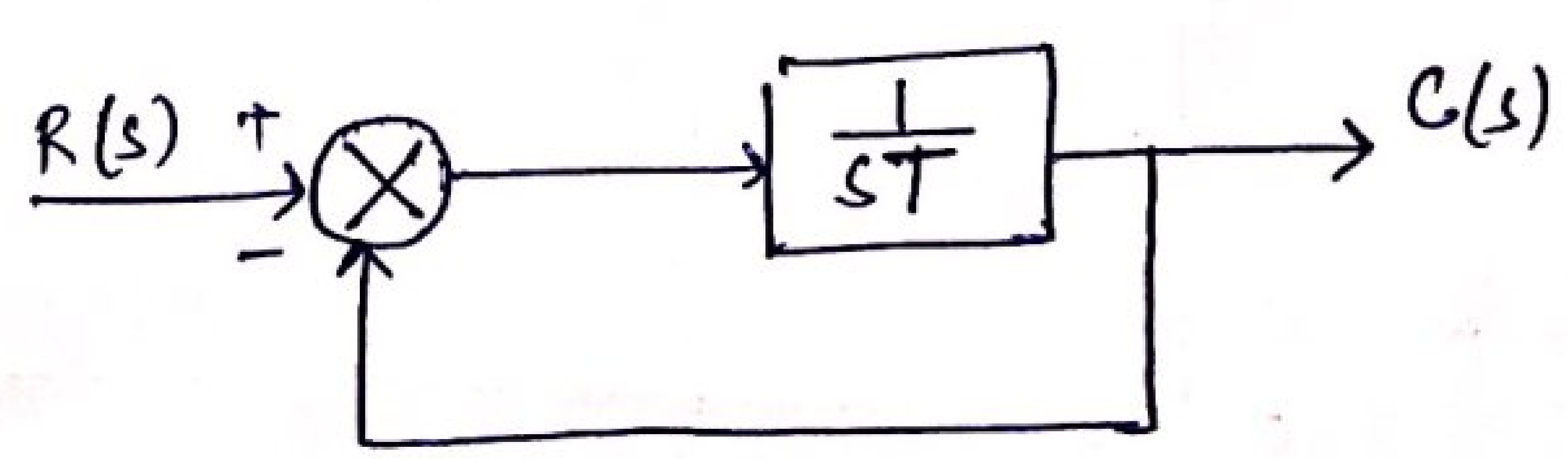
A control system in which the highest power of s in the denominator of its transfer func^n is equal to 1, is called the first order control system.

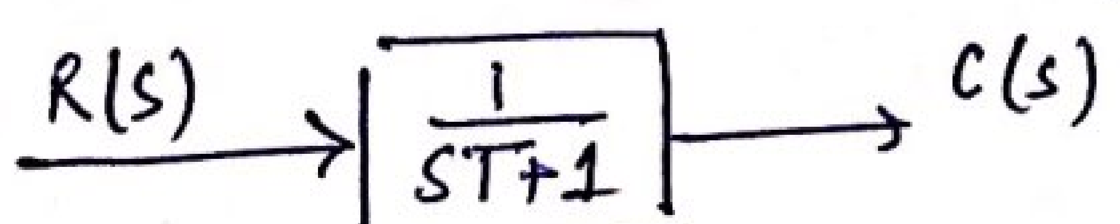
Consider a 1st order control system with unity feedback,

$$G(s) = \frac{1}{sT}, \quad H(s) = 1$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} = \frac{\frac{1}{sT}}{1 + \frac{1}{sT}\cdot 1}$$

$$\frac{C(s)}{R(s)} = \frac{1}{sT+1}.$$





Response of 1st Order System with Unit Step Input

For 1st Order System,

$$\frac{C(s)}{R(s)} = \frac{1}{sT+1} \quad \text{——— (1)}$$

$$C(s) = \frac{1}{sT+1} R(s)$$

$R(s) = \frac{1}{s}$ {Since Laplace transform of unit step funcn is $1/s$}

$$C(s) = \frac{1}{sT+1} \cdot \frac{1}{s} \quad \text{——— (2)}$$

Expanding $C(s)$ in partial fractions

$$C(s) = \frac{1}{s} - \frac{T}{1+sT}.$$

$$C(s) = \frac{1}{s} - \frac{1}{s+(1/T)} \quad \text{——— (3)}$$

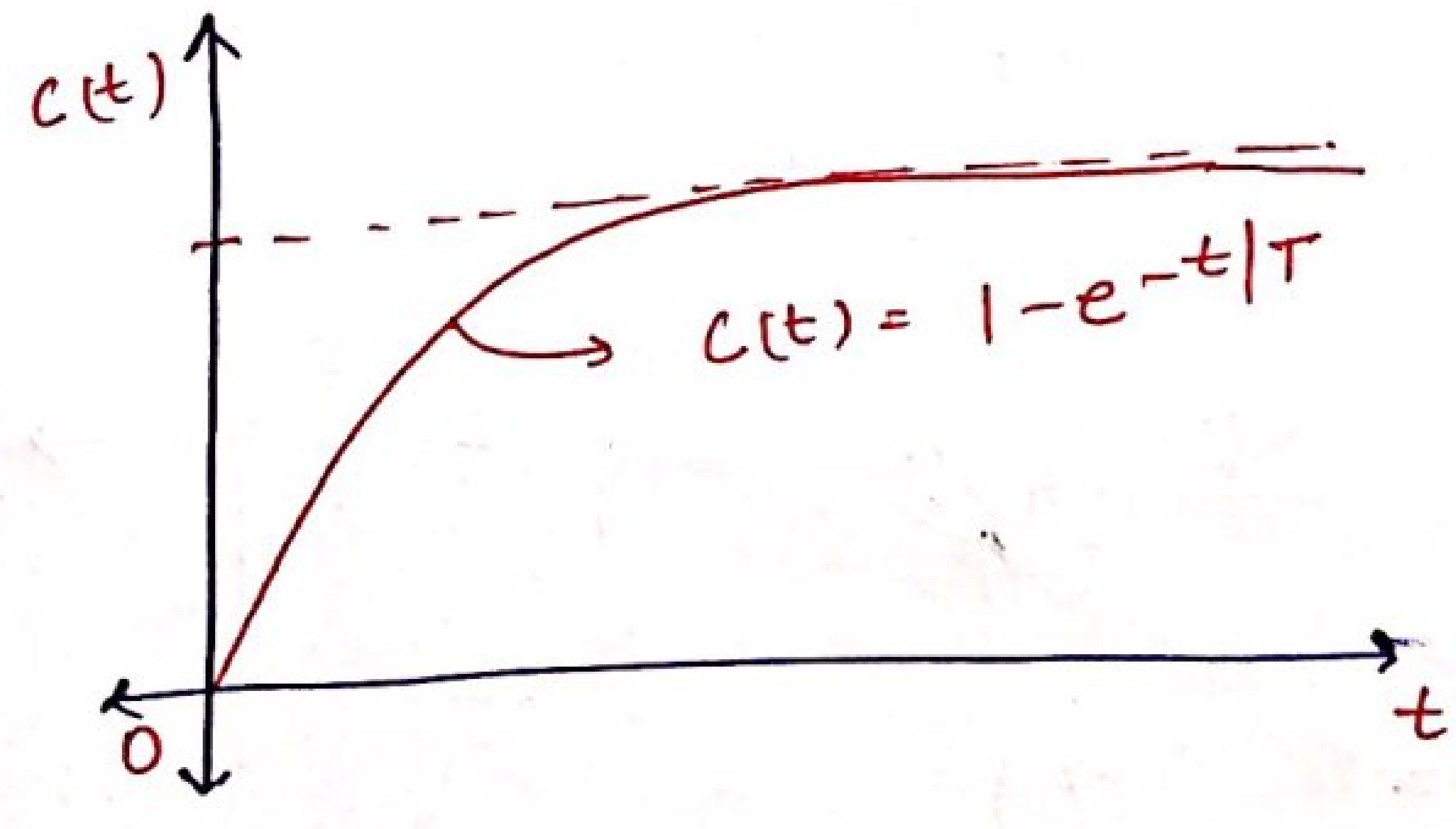
Taking inverse laplace transform of eqn (3)

$$c(t) = 1-e^{-t/T} \text{ (for } t \geq 0)$$

at $t=0$, $c(t)=0$

$t=\infty$, $c(t)=1$.

$t=T$, $c(t) = 1-e^{-1} = 0.632$



Response of the First-Order System with Unit Ramp funcⁿ :-

For first Order System

$$\frac{C(s)}{R(s)} = \frac{1}{sT+1}$$

$$C(s) = R(s) \cdot \frac{1}{sT+1}$$

(Since Laplace transform of Unit ramp is $1/s^2$)

$$C(s) = \frac{1}{s^2} \cdot \frac{1}{sT+1}$$

Expanding $C(s)$ into partial fractions

$$C(s) = \frac{1}{s^2} - \frac{T}{s} + \frac{T^2}{sT+1}$$

$$C(s) = \frac{1}{s^2} - \frac{T}{s} + T \cdot \frac{1}{s+\frac{1}{T}}$$

Taking inverse laplace transform of above eqⁿ :-

$$c(t) = t - T + Te^{-t/T} \quad (\text{for } t \geq 0)$$

The error signal $e(t) = r(t) - c(t)$

$$= t - [t - T + Te^{-t/T}]$$

$$e(t) = T(1 - e^{-t/T})$$

Steady state error, $e_{ss} = \lim_{t\to\infty} e(t)$

$$= \lim_{t\to\infty} T(1 - e^{-t/T})$$

$$= T - 0 = T$$

for the unit - impulse input,

$$R(s) = 1$$

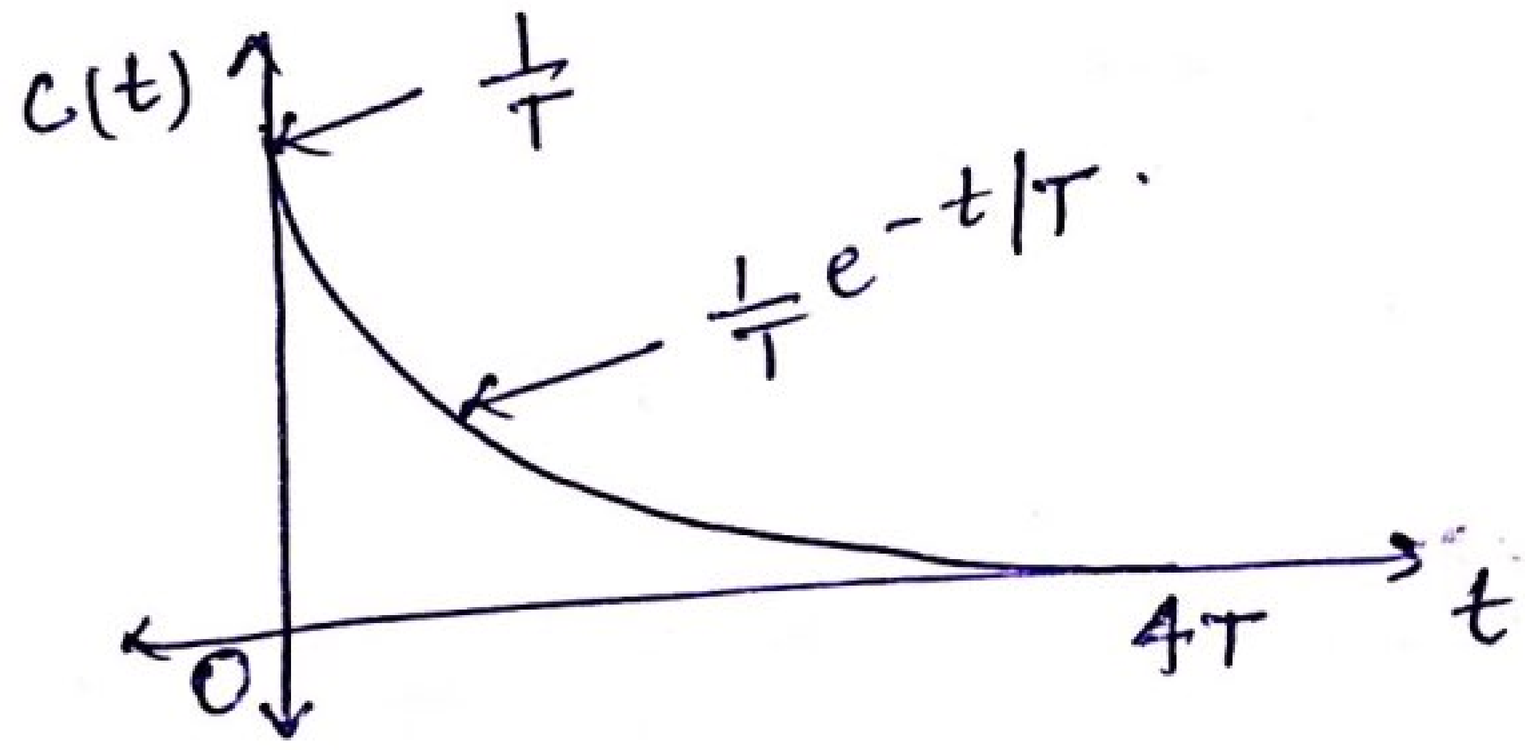
$$C(s) = \frac{1}{sT+1} \cdot R(s)$$

$$\therefore C(s) = \frac{1}{sT+1} \cdot 1.$$

$$C(s) = \frac{1}{T} \cdot \frac{1}{s+\frac{1}{T}}$$

Taking inverse L.T. of above eq^n:-

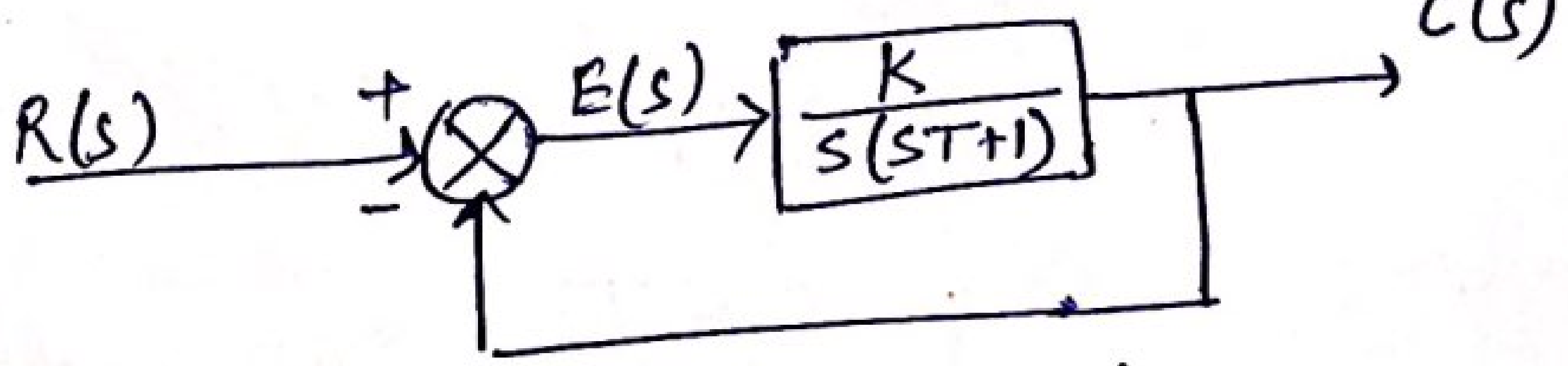
$$c(t) = \frac{1}{T} e^{-t/T}, \text{ for } t \geq 0$$



RESPONSE OF A SECOND - ORDER SYSTEM

A control system in which the highest power of s in the denominator of in the transfer func^n is equal to 2 is called the 2nd Order control system.

$$\text{Here } G(s) = \frac{K}{s(sT+1)}, \quad H(s) = 1$$



$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{K/T}{s^2+\frac{1}{T}s+\frac{K}{T}} \quad \text{—— (1)}$$

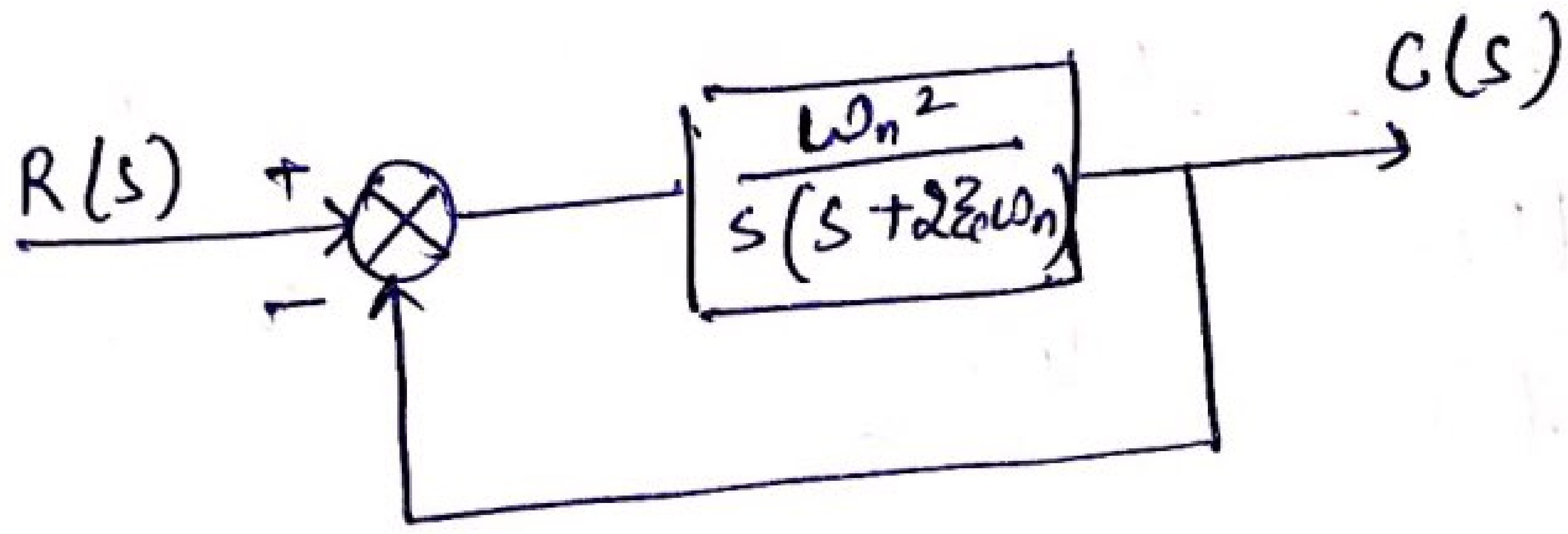
$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \quad \text{——— (2)}$$

where, $\omega_n = \sqrt{\frac{K}{T}} \triangleq$ undamped natural freq.

and. $\xi = \frac{1}{2\omega_n T} \triangleq$ damping ratio or damping factor

Characteristic Equation :- $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$



General form of characteristic eqn is :-

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

Solving the above eqn, we get

$$s_1 = -\xi\omega_n + \omega_n\sqrt{\xi^2 - 1}$$

$$s_2 = -\xi\omega_n - \omega_n\sqrt{\xi^2 - 1}$$

These roots are the poles of transfer funcn. The nature of the roots of the characteristic eqn indicate the time response. $\frac{1}{\xi\omega_n}$ is the time constant of the system.

(i) Case 1 :- $(\xi < 1)$ (s_1 and s_2 are complex)

(Damping freq :- $\omega_d = \omega_n\sqrt{1 - \xi^2}$)

$$s_1 = -\xi\omega_n + j\omega_n\sqrt{1 - \xi^2}$$

$$s_2 = -\xi\omega_n - j\omega_n\sqrt{1 - \xi^2}$$

$$\therefore s_1 = -\xi\omega_n + j\omega_d$$

$$s_2 = -\xi\omega_n - j\omega_d$$

* We consider the 4 cases

(i) $0 < \xi_e < 1$ (Underdamped case)

(ii) $\xi_e = 1$ (Critically Damped)

(iii) $\xi_e > 1$ (Overdamped)

(iv) $\xi_e = 0$ (Undamped case)

1.) Underdamped Case :- $(0 < \xi_e < 1)$

Let us now solve for the response of the system to the unit-step input

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi_e \omega_n s + \omega_n^2}$$

$$C(s) = \frac{\omega_n^2}{s^2 + 2\xi_e \omega_n s + \omega_n^2} \cdot \left(\frac{1}{s}\right)$$

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\xi_e \omega_n s + \omega_n^2)}$$

Replace $s^2 + 2\xi_e \omega_n s + \omega_n^2$ by $(s + \xi_e \omega_n)^2 + \omega_n^2(1-\xi_e^2)$

$$C(s) = \frac{\omega_n^2}{s((s + \xi_e \omega_n)^2 + \omega_n^2(1-\xi_e^2))}$$

$$= \frac{A}{s} + \frac{B}{(s + \xi_e \omega_n)^2 + \omega_d^2}$$

$A = 1$ and $B = -(s + 2\xi_e \omega_n)$

$$\therefore C(s) = \frac{1}{s} - \frac{s + 2\xi_e \omega_n}{(s + \xi_e \omega_n)^2 + \omega_d^2}$$

$$= \frac{1}{s} - \frac{s + \xi_e \omega_n + \xi_e \omega_n}{(s + \xi_e \omega_n)^2 + \omega_d^2}$$

$$C(s) = \frac{1}{s} - \left\{ \frac{s+\xi_e \omega_n}{(s+\xi_e\omega_n)^2+\omega_d^2} + \frac{\xi_e \omega_n}{\omega_d} \cdot \frac{\omega_d}{(s+\xi_e\omega_n)^2+\omega_d^2} \right\}$$

$$\therefore \; c(t) = 1 - \left[e^{-\xi_e \omega_n t} \cdot \cos \omega_d t + \frac{\xi_e \omega_n}{\omega_d} e^{-\xi_e \omega_n t} \cdot \sin \omega_d t \right]$$

Putting $\omega_d = \omega_n \sqrt{1-\xi_e^2}$

$$c(t) = 1 - e^{-\xi_e \omega_n t} \left[\cos \omega_d t + \frac{\xi_e}{\sqrt{1-\xi_e^2}} \sin \omega_d t \right]$$

$$= 1 - \frac{e^{-\xi_e \omega_n t}}{\sqrt{1-\xi_e^2}} \left[\sqrt{1-\xi_e^2} \cdot \cos \omega_d t + \xi_e \sin \omega_d t \right]$$

Putting $\sqrt{1-\xi_e^2} = \sin \phi$

$\cos \phi = \xi_e, \quad \tan \phi = \frac{\sqrt{1-\xi_e^2}}{\xi_e}$

$$\therefore \; c(t) = 1 - \frac{e^{-\xi_e \omega_n t}}{\sqrt{1-\xi_e^2}} \left[\sin \phi \cdot \cos \omega_d t + \cos \phi \sin \omega_d t \right]$$

$$= 1 - \frac{e^{-\xi_e \omega_n t}}{\sqrt{1-\xi_e^2}} \sin (\omega_d t + \phi)$$

$\therefore \; \omega_d = \omega_n \sqrt{1-\xi_e^2}$ and $\phi = \tan^{-1} \frac{\sqrt{1-\xi_e^2}}{\xi_e}.$

$$\Rightarrow \; c(t) = 1 - \frac{e^{-\xi_e \omega_n t}}{\sqrt{1-\xi_e^2}} \sin \left[(\omega_n \sqrt{1-\xi_e^2})\, t + \tan^{-1} \frac{\sqrt{1-\xi_e^2}}{\xi_e} \right]$$

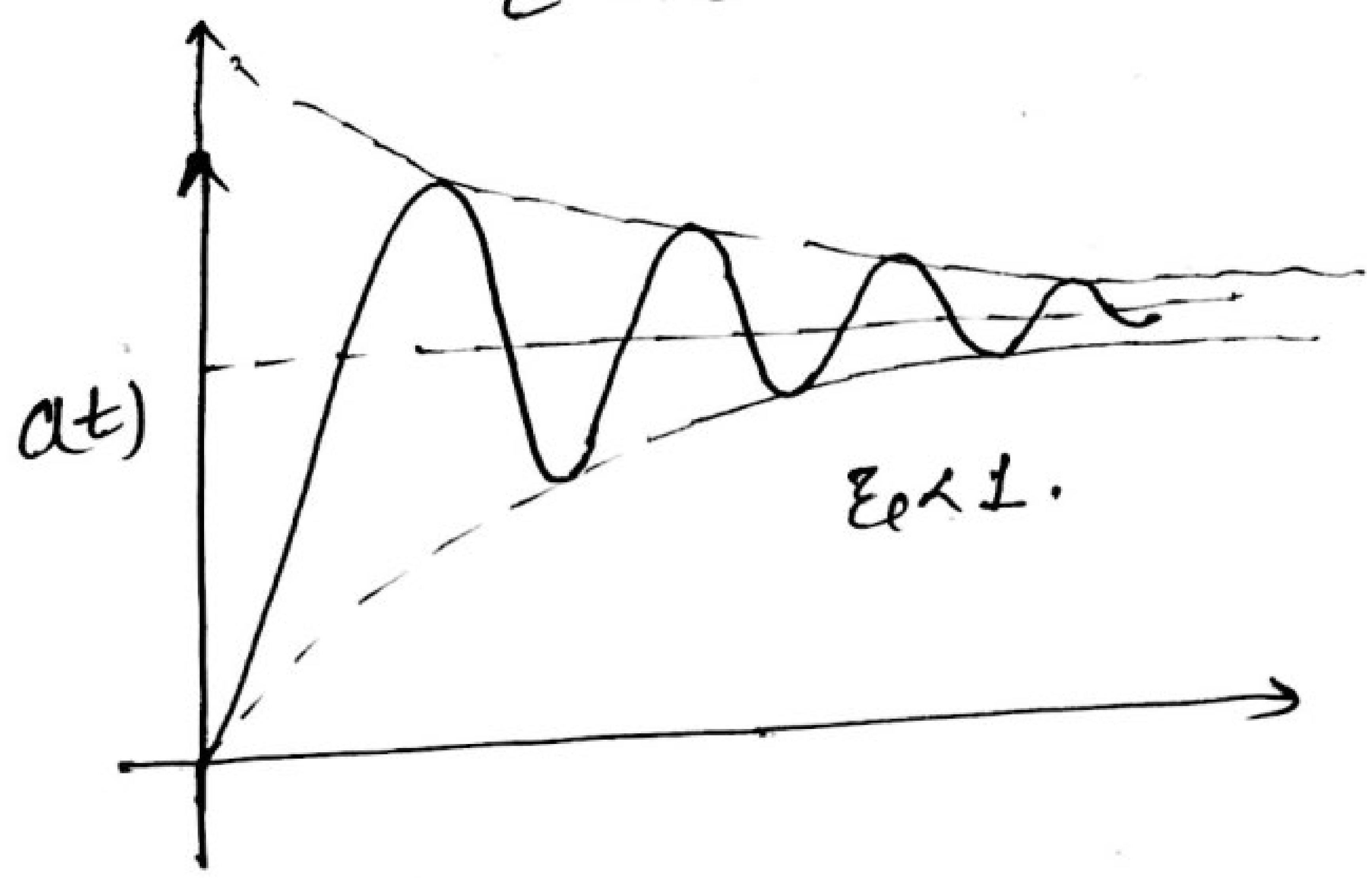
The error signal for the system

$e(t) = r(t) - c(t)$

$$\therefore \quad c(t) = \frac{e^{-\xi_e \omega_n t}}{\sqrt{1-\xi_e^2}} \sin\left[(\omega_n \sqrt{1-\xi_e^2})\, t + \tan^{-1}\frac{\sqrt{1-\xi_e^2}}{\xi_e}\right]$$

The steady state value of $c(t)$ is

$$c_{ss} = \lim_{t \to \infty} c(t) = 1.$$



2. Critically Damped case :- ($\xi_e = 1$)

If the two poles of $\frac{C(s)}{R(s)}$, are equal, the system is said to be a critically damped one

Subsituting $\xi_e = 1$ in

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s^2 + 2\omega_n s + \omega_n^2)}$$

for step input, $R(s) = \frac{1}{s}$

$$C(s) = \frac{\omega_n^2}{s(s+\omega_n)^2}$$

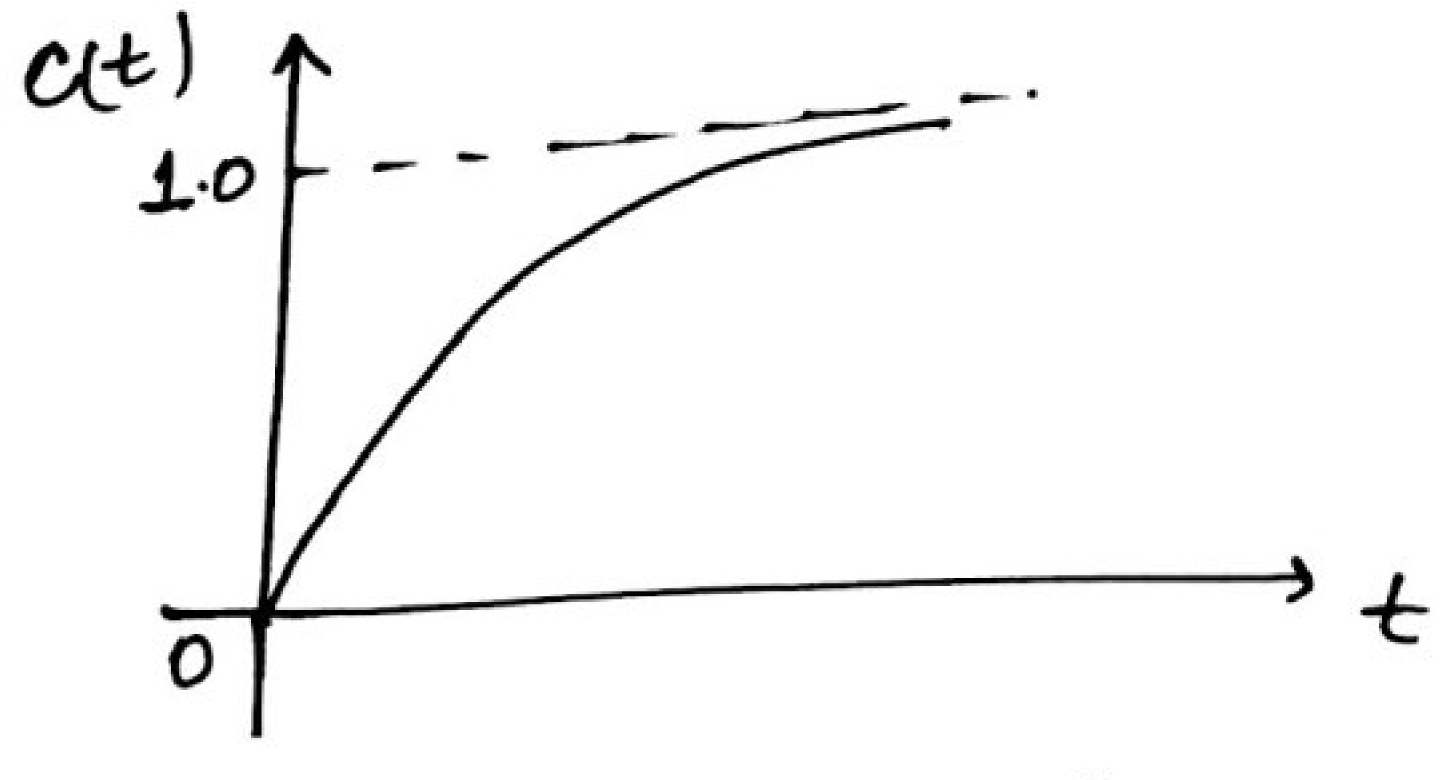
Resolving into partial fraction

$$C(s) = \frac{1}{s} - \frac{\omega_n}{(s+\omega_n)^2} - \frac{1}{s+\omega_n}$$

Taking the inverse laplace transform, we get.

$$c(t) = 1 - \omega_n t e^{-\omega_n t} - e^{-\omega_n t}$$

$$c(t) = 1 - (1 + \omega_n t) e^{-\omega_n t}; \quad t \geq 0$$



3. Undamped case ($\xi_e = 0$)

Subsituting $\xi_e = 0$ in eq^n $\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi_e\omega_n s + \omega_n^2}$

we get; $C(s) = \frac{\omega_n^2}{s^2 + \omega_n^2} R(s)$

$$\Rightarrow C(s) = \frac{\omega_n^2}{s(s^2 + \omega_n^2)} = \frac{1}{s} - \frac{s}{s^2 + \omega_n^2}$$

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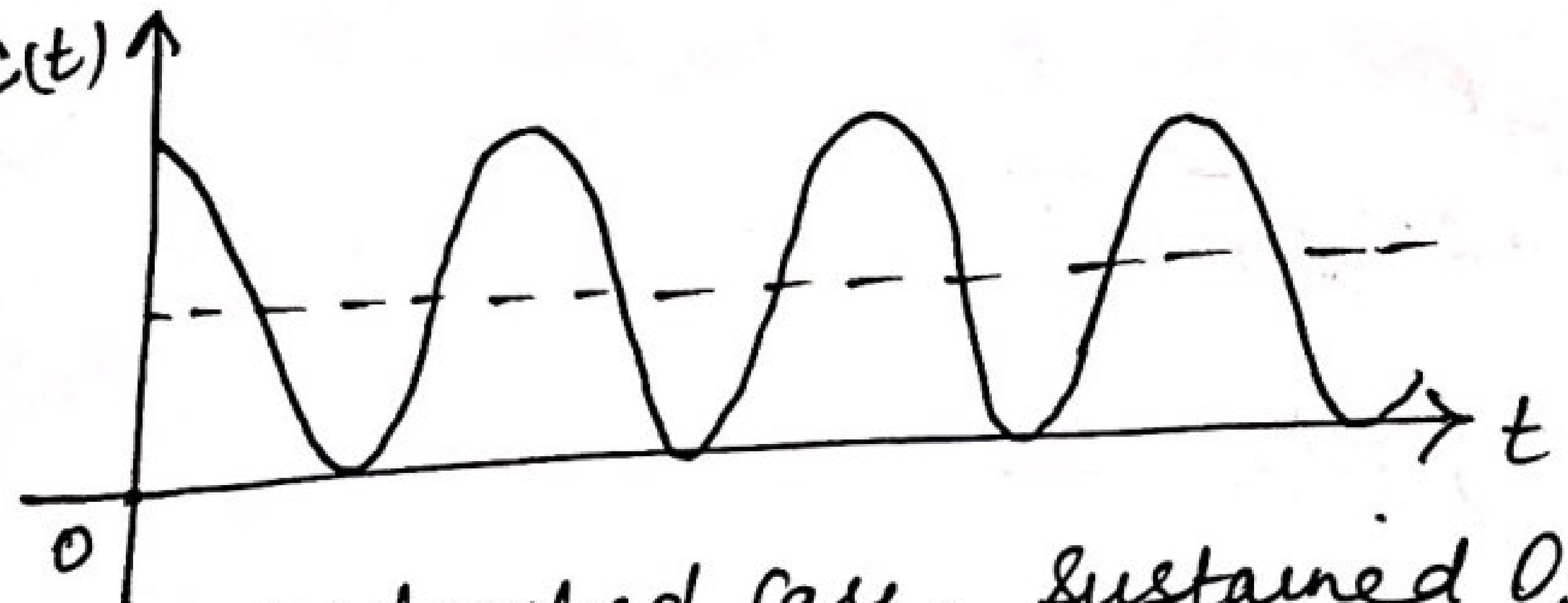
$$\therefore \; c(t) = 1 - \cos\omega_n t$$

Since there is no time damping, this response doesnot die out with time. This response is Called the undamped response



undamped case, Sustained Oscillations.

4.7 $\xi_e > 1$, Overdamped case :-

for $\xi_e > 1$,

$$C(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{(s+\xi_e\omega_n)^2 - \omega_n^2(\xi_e^2-1)}$$

put $\omega_d^2 = \omega_n^2(\xi_e^2-1)$

$$\therefore \; C(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{(s+\xi_e\omega_n)^2 - \omega_d^2}$$

$$C(s) = \frac{1\omega_n^2}{s(s+\xi_e\omega_n-\omega_d)(s+\xi_e\omega_n+\omega_d)}$$

$$= \frac{A}{s} + \frac{B}{s+\xi_e\omega_n+\omega_d} + \frac{C}{s+\xi_e\omega_n-\omega_d}$$

we get $A = 1$

$$B = \frac{\omega_n^2}{(\xi_e\omega_n+\omega_d)(2\omega_d)}$$

Putting $\omega_d^2 = \omega_n^2(\xi_e^2-1)$ and Simplify for B :-

$$B = \frac{1}{2\sqrt{\xi_e^2-1}\left[\xi_e+\sqrt{\xi_e^2-1}\right]}$$

Similarly, $C = \dfrac{-1}{2\sqrt{\xi_e^2-1}\left[\xi_e-\sqrt{\xi_e^2-1}\right]}$

$$C(s) = \frac{1}{s} + \frac{1}{2\sqrt{\xi_e^2-1}\left[\xi_e+\sqrt{\xi_e^2-1}\right](s+\xi_e\omega_n+\omega_d)} - \frac{1}{2\sqrt{\xi_e^2-1}\left[\xi_e-\sqrt{\xi_e^2-1}\right](s+\xi_e\omega_n-\omega_d)}$$

Put the value of ω_d

$$C(s) = \frac{1}{s} + \frac{1}{2\sqrt{\xi_e^2-1}\left[\xi_e+\sqrt{\xi_e^2-1}\right]\left[s+\xi_e\omega_n+\omega_n\sqrt{\xi_e^2-1}\right]} - \frac{1}{2\sqrt{\xi_e^2-1}\left[\xi_e-\sqrt{\xi_e^2-1}\right]\left(s+\xi_e\omega_n-\omega_n\sqrt{\xi_e^2-1}\right)}$$

$$C(s) = \frac{1}{s} + \frac{1}{2\sqrt{\xi_e^2-1}\left[\xi_e+\sqrt{\xi_e^2-1}\right]\left[s+\omega_n\left[\xi_e+\sqrt{\xi_e^2-1}\right]\right]} - \frac{1}{2\sqrt{\xi_e^2-1}\left[\xi_e-\sqrt{\xi_e^2-1}\right]\left[s+\omega_n\left[\xi_e-\sqrt{\xi_e^2-1}\right]\right]}$$

Inverse laplace transform of above eq^n is

$$C(t) = 1 + \frac{e^{-\left(\xi_e+\sqrt{\xi_e^2-1}\right)\omega_n t}}{2\sqrt{\xi_e^2-1}\left(\xi_e+\sqrt{\xi_e^2-1}\right)} - \frac{e^{-\left(\xi_e-\sqrt{\xi_e^2-1}\right)\omega_n t}}{2\sqrt{\xi_e^2-1}\left(\xi_e-\sqrt{\xi_e^2-1}\right)}$$

from above eq^n, we get two time constants

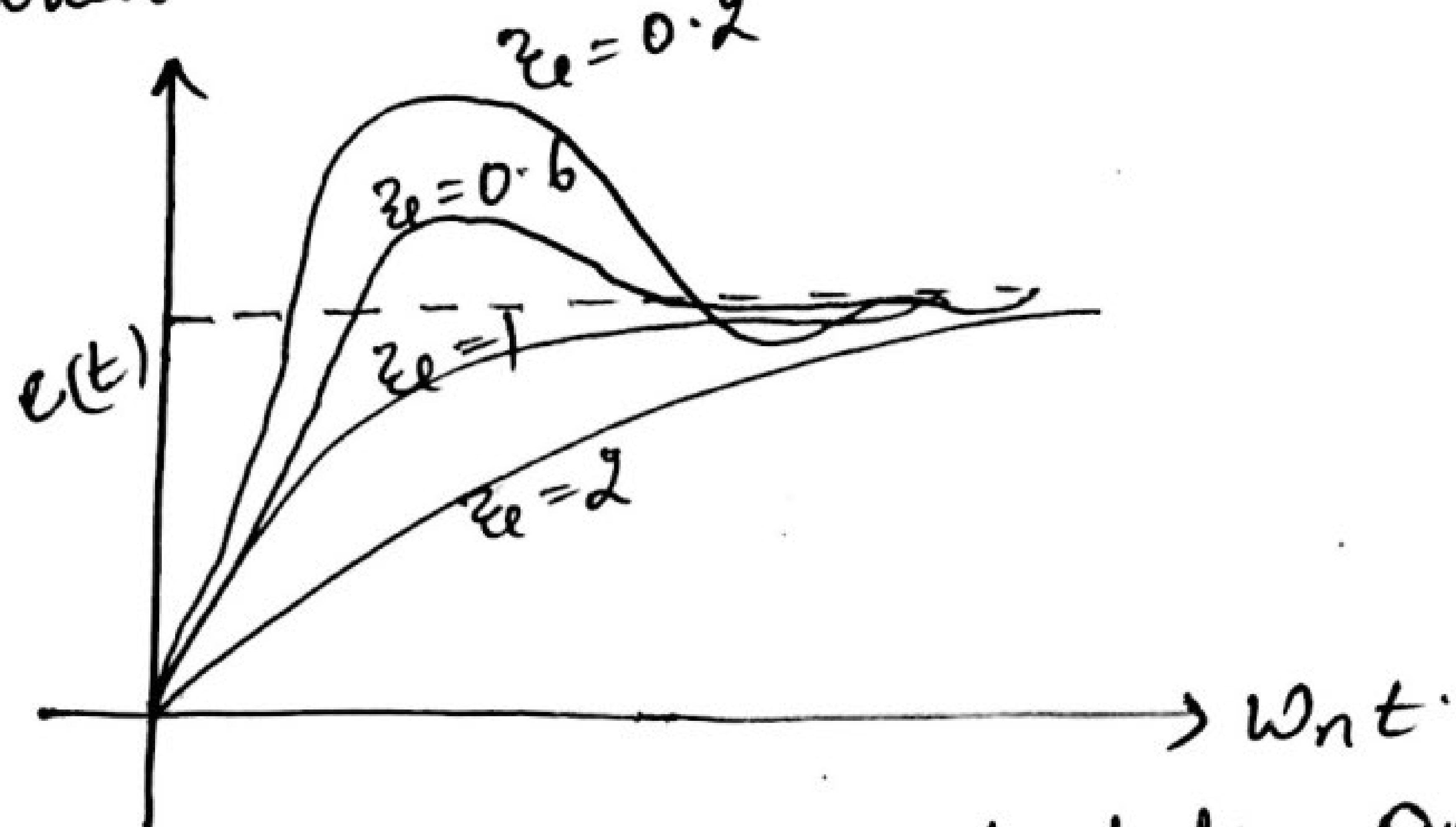
$$T_1 = \frac{1}{\left(\xi_e+\sqrt{\xi_e^2-1}\right)\omega_n} \quad ; \quad T_2 = \frac{1}{\left(\xi_e-\sqrt{\xi_e^2-1}\right)\omega_n}$$

We observe that when ξ_e is greater than one, there are two exponential term, the first term has a time constant T_1 which is smaller than the time constant of other exponential term. In other words we can say that first exponential term decaying much faster than other exponential term so, we can neglect the first exponential term.

$$c(t) = 1 - \frac{e^{-(\xi - \sqrt{\xi^2 - 1})\omega_n t}}{2\sqrt{\xi^2 - 1}\,(\xi - \sqrt{\xi^2 - 1})}$$

and time constant $= \dfrac{1}{(\xi - \sqrt{\xi^2 - 1})\omega_n}$

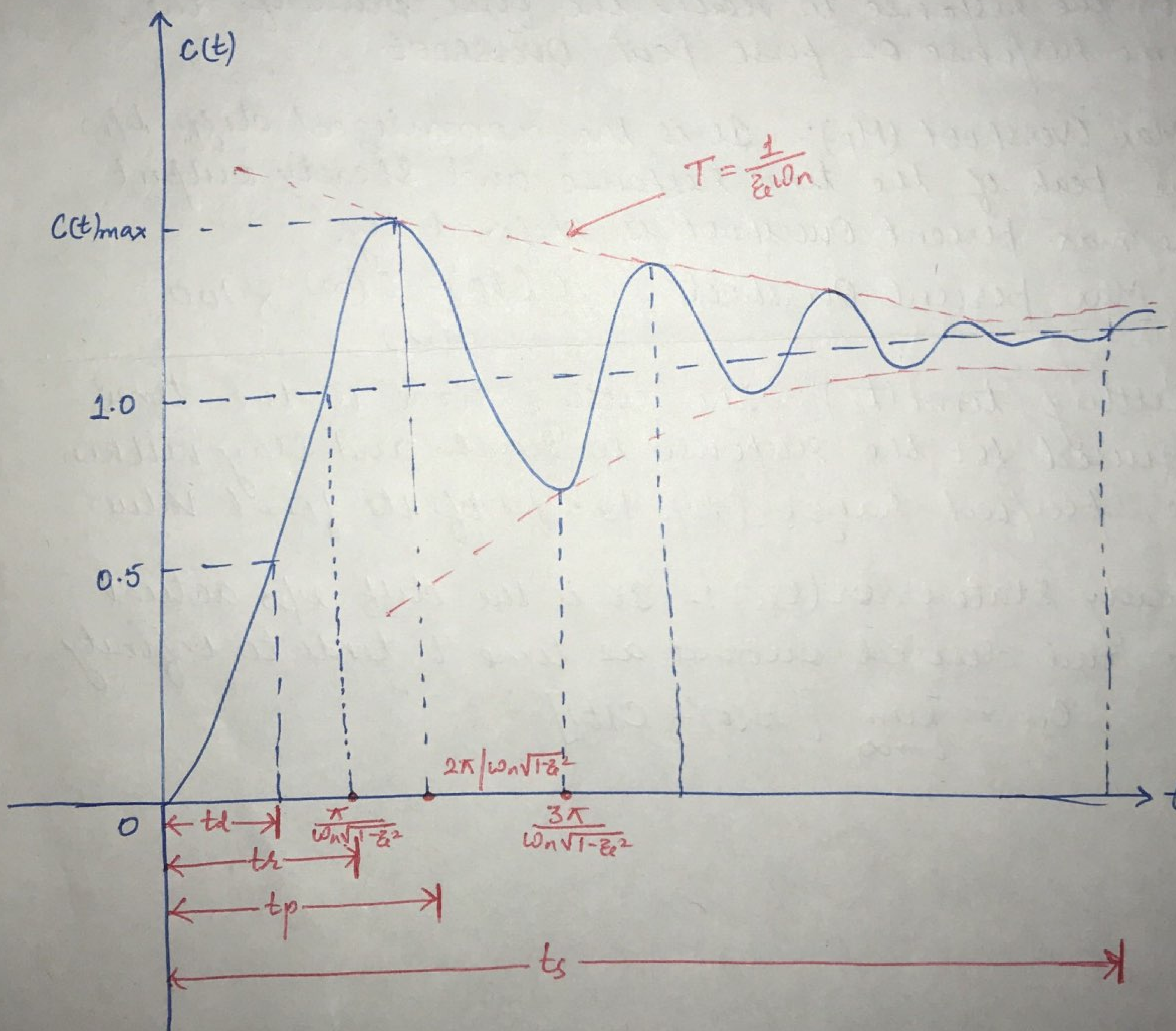
for different values of ξ, the curves of $c(t)$ is shown.



From the curve it is clear that the overdamped systems are sluggish.

TRANSIENT RESPONSE SPECIFICATIONS OF SECOND ORDER SYSTEM

The transient response of a control system to a unit step i/p depends upon the intial cond's. consider a 2nd order system with unit step input and the system intially at rest i.e. all intial cond's are zero. The following are the common transient response characterstics :-



1) Delay Time (t_d) : The delay time is the time required for the response to reach 50% of the final Value in first time.

2) Rise time (t_r) :- It is the time required for the responce to rise from 10% to 90% of its final Value for overdamped Systems and 0 to 100% for underdamped Systems.

3) Peak time (t_p) :- The peak time is the time required for the responce to reach the first peak of the time response or first peak Overshoot

4) Max. Overshoot (M_p):- It is the normalized diff. b/w the peak of the time responce and Steady output The max. percent Overshoot is defined by

$$\text{Max. percent Overshoot} = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100$$

5) Settling time (t_s) : The Settling time is the time required for the responce to reach and Stay within the specified range (2% to 5%) of its final Value.

6) Steady State error (e_{ss}) :- It is the diff b/w actual o/p and desired output as time 't' tends to infinity

$$e_{ss} = \lim_{t \to \infty} [r(t) - c(t)]$$

Expression for Rise time (t_r) :-

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[(\omega_n\sqrt{1-\xi^2})t + \phi\right]$$

where $\phi = \tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}$

Let the response reaches 100% of desired value.
Put $c(t) = 1$

$$1 = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[(\omega_n\sqrt{1-\xi^2})t + \phi\right]$$

$$\frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[(\omega_n\sqrt{1-\xi^2})t + \phi\right] = 0$$

Since $e^{-\xi\omega_n t} \neq 0$

$\therefore \sin\left[(\omega_n\sqrt{1-\xi^2})t + \phi\right] = 0$

$\sin\left[(\omega_n\sqrt{1-\xi^2})t + \phi\right] = \sin n\pi$

Put $n = 1$.

$$(\omega_n\sqrt{1-\xi^2})t_r + \phi = \pi$$

$$t_r = \frac{\pi - \phi}{\omega_n\sqrt{1-\xi^2}}$$

$$\boxed{t_r = \frac{\pi - \tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}}{\omega_n\sqrt{1-\xi^2}}}$$

Expression for Peak time :-

Since $c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[(\omega_n\sqrt{1-\xi^2})t + \phi\right]$

For maximum $\frac{dc(t)}{dt} = 0$

$$\frac{dc(t)}{dt} = 0$$

$$\frac{dc(t)}{dt} = \frac{-e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \cos\left[(\omega_n\sqrt{1-\xi^2})t+\phi\right] \omega_n\sqrt{1-\xi^2} + \sin\left[(\omega_n\sqrt{1-\xi^2})t+\phi\right] e^{-\xi\omega_n t} \frac{\xi\omega_n}{\sqrt{1-\xi^2}}$$

Since; $e^{-\xi\omega_n t} \neq 0$

$$\cos\left[(\omega_n\sqrt{1-\xi^2})t+\phi\right]\sqrt{1-\xi^2} = \sin\left[(\omega_n\sqrt{1-\xi^2})t+\phi\right]\xi$$

Putting $\sqrt{1-\xi^2} = \sin\phi$ and $\xi = \cos\phi$

$$\therefore \frac{\sin\left[(\omega_n\sqrt{1-\xi^2})t+\phi\right]}{\cos\left[(\omega_n\sqrt{1-\xi^2})t+\phi\right]} = \frac{\sin\phi}{\cos\phi}$$

$$\tan\left[(\omega_n\sqrt{1-\xi^2})t+\phi\right] = \tan\phi.$$

the time to various peaks

$$(\omega_n\sqrt{1-\xi^2})t_p = n\pi$$

where $n = 0, 1, 2, 3, \ldots$

Max. Overshoot identified by putting $n = 1$, therefore the peak time to the 1st Overshoot.

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}} \quad \text{or} \quad \omega_n\sqrt{1-\xi^2}\, t + \phi = n\pi + \phi$$

$$\boxed{t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}}$$

The 1st min. (undershoot) occurs at $n = 2$

$$\omega_n\sqrt{1-\xi^2}\, t = n\pi$$

$$t_p = \frac{n\pi}{\omega_n\sqrt{1-\xi^2}}$$

$$t_{min} = \frac{2\pi}{\omega_n\sqrt{1-\xi^2}}$$

Expression for Max. Overshoot M_p :-

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[(\omega_n\sqrt{1-\xi^2})t + \phi\right]$$

Max. overshoot occurs at $t = t_p$

Put, $t = t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}$ in above eqn

$$c(t) = 1 - \frac{e^{-\xi\omega_n \frac{\pi}{\omega_n\sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}} \sin\left[\omega_n\sqrt{1-\xi^2} \cdot \frac{\pi}{\omega_n\sqrt{1-\xi^2}} + \phi\right]$$

$$= 1 - \frac{e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}} \sin(\pi + \phi)$$

$$= 1 + \frac{e^{-\pi\xi/\sqrt{1-\xi^2}}}{\sqrt{1-\xi^2}} \sin\phi$$

Put $\sin\phi = \sqrt{1-\xi^2}$

$$c(t)_{max} = 1 + e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}}$$

$$M_p = c(t)_{max} - 1.$$

$$M_p = e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}}$$

$$\%M_p = e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}} \times 100.$$

Settling time (t_s) :-

The time constant of the curve is $\frac{1}{\xi\omega_n}$. The speed of the decay depends upon the time constant. The settling time for a second order System is approximately 4 times the time constant ($1/\xi\omega_n$)

$$\therefore \; t_s = \frac{4}{\xi\omega_n}$$

Ques. When a 2nd Order control system is subjected to a unit step input, the values of $\xi = 0.5$ and $\omega_n = 6$ rad/sec. Determine rise time, peak time, settling time and peak overshoot.

$\xi = 0.5$
$\omega_n = 6$ rad/s

1.) $t_r = \dfrac{\pi - \tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}}{\omega_n\sqrt{1-\xi^2}} = \dfrac{3.14 - 1.047}{5.19} = 0.403$ sec.

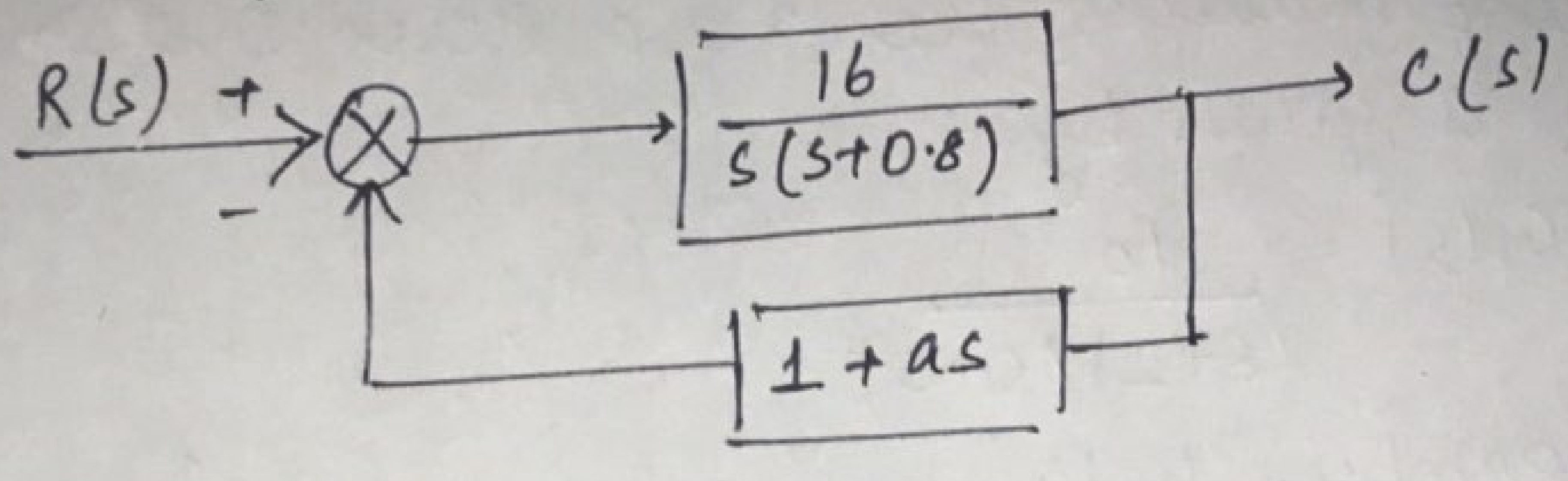
2.) $t_p = \dfrac{\pi}{\omega_n\sqrt{1-\xi^2}} = \dfrac{\pi}{6\sqrt{1-(0.5)^2}} = 0.605$ sec.

3.) Settling time $t_s = \dfrac{4}{\xi\omega_n} = \dfrac{4}{0.5\times 6} = 1.33$ sec.

4.) Max. Overshoot, $M_p = e^{-\pi\xi/\sqrt{1-\xi^2}} \times 100$

$= 0.163 \times 100 = 16.3\%$

Ques1. Consider the system as shown in figure. Determine the value of 'a' such that the damping ratio is 0.5. Also obtain the values of rise time and max. overshoot in its step response.



Solution:-

$$\frac{C(s)}{R(s)} = \frac{\dfrac{16}{s(s+0.8)}}{1 + \dfrac{16}{s(s+0.8)}(1+as)}$$

$$\frac{C(s)}{R(s)} = \frac{16}{s^2 + (0.8 + 16a)s + 16} \quad \text{——— (1)}$$

The characteristic eq^n is $s^2 + (0.8 + 16a)s + 16 = 0$ ——— (2)

comparing eq^n (2) by:- $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

$2\xi\omega_n = 0.8 + 16a$

$\omega_n^2 = 16$

$\omega_n = 4$

$8\xi = 0.8 + 16a$, ($\xi = 0.5$; given)

$4 - 0.8 = 16a$

$16a = 3.2$

$a = \dfrac{3.2}{16} = 0.2$

$\boxed{a = 0.2}$

Rise time $t_r = \dfrac{\pi - \tan^{-1}\dfrac{\sqrt{1-\xi^2}}{\xi}}{\omega_n\sqrt{1-\xi^2}} = 0.605$ sec. Ans

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Max. Overshoot, $M_p = e^{-\pi\xi/\sqrt{1-\xi^2}} \times 100$

$= e^{-\pi(0.5)/\sqrt{1-(0.5)^2}} \times 100$

$= 16.31\%$.

Ques 2. The open loop transfer funcⁿ of a servo system with unity feedback is given by:-

$$G(s) = \frac{10}{(s+2)(s+5)}$$

Determine the damping ratio, undamped natural freq., what is the % overshoot of the response to a unit step input.

Solution:- $G(s) = \frac{10}{(s+2)(s+5)}$

$H(s) = 1$

Characteristic Eqⁿ:- $1 + G(s)H(s) = 0$

$$1 + \frac{10}{(s+2)(s+5)} = 0$$

$$s^2 + 7s + 20 = 0$$

Compare with $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

$\omega_n^2 = 20$

$2\xi\omega_n = 7$

$$\boxed{\begin{aligned} \omega_n &= 4.472 \text{ rad/sec} \\ \xi &= 0.7826 \end{aligned}}$$

$M_p = e^{-\pi\xi/\sqrt{1-\xi^2}} \times 100$

$$\boxed{M_p = 1.92\%}$$

Ques 3. Measurement conducted on a servo mechanism

Show the system response to be:-

$$c(t) = 1 + 0.2e^{-60t} - 1.2e^{-10t}$$

when subjected to a unit step input.

(a) Obtain the expression for closed loop transfer funcⁿ

(b) Determine the undamped natural freq. and damping ratio of the system

solution:- $c(t) = 1 + 0.2e^{-60t} - 1.2e^{-10t}$

$$C(s) = \frac{1}{s} + \frac{0.2}{s+60} - \frac{1.2}{(s+10)}$$

$$\therefore C(s) = \frac{600}{s(s+60)(s+10)}$$

(a) Transfer funcⁿ

$$\frac{C(s)}{R(s)} = \frac{600}{s(s+60)(s+10)}$$

$$C(s) = \frac{600}{(s+60)(s+10)}$$

(b) characteristic eqⁿ:-

$$s^2 + 70s + 600 = 0 \quad \text{—— (1)}$$

comparing eqⁿ (1) with $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

$$2\xi\omega_n = 70$$

$$\omega_n^2 = 600$$

$$\therefore \omega_n = 24.49 \text{ rad/sec. Ans}$$

$$2 \times \xi \times 24.49 = 600$$

$$\xi = 1.42 \text{ Ans.}$$

Ques 4. Consider the unit-step response of a unity feedback control system whose open loop transfer funcⁿ is

$$G(s) = \frac{1}{s(s+1)}$$

Obtain its rise time, peak time, max. overshoot & settling time

Characteristic eqn:- $1+G(s)H(s)=0$

$$1+\frac{1}{s(s+1)}\cdot 1=0$$

$$s^2+s+1=0 \quad \text{—— (1)}$$

Compare with $s^2+2\xi\omega_n s+\omega_n^2=0$

$\omega_n^2=1$; $\omega_n=1$ rad/sec

$2\xi\omega_n=1$; $\therefore \xi=0.5$

* $t_r = \dfrac{\pi-\tan^{-1}\dfrac{\sqrt{1-\xi^2}}{\xi}}{\omega_n\sqrt{1-\xi^2}} = 2.41$ sec

* Peak time $t_p = \dfrac{\pi}{\omega_n\sqrt{1-\xi^2}} = \dfrac{\pi}{1\sqrt{1-0.5^2}} = 3.625$ sec.

* $M_p = e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}}\times 100$

$= 16.31\%$

* $t_s = \dfrac{4}{\xi\omega_n} = \dfrac{4}{0.5\times 1} = 8$ sec.

Ques. Obtain the unit step response for unity feedback system whose open loop transfer func^n is

$$G(s) = \frac{4}{s(s+5)}$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

$$\frac{C(s)}{R(s)} = \frac{4}{s^2+5s+4}$$

$$C(s) = \frac{4}{s(s^2+5s+4)} = \frac{4}{s(s+1)(s+4)}$$

- Resolving the above eq^n into partial fraction we get,

$$C(s) = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+4}$$

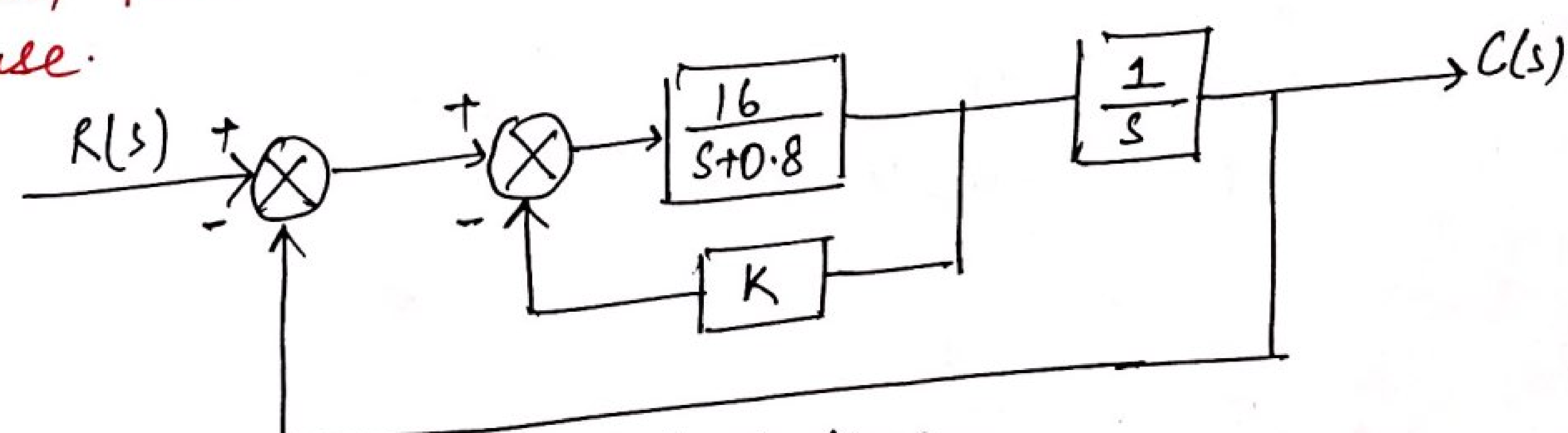
$$C(s) = \frac{1}{s} - \frac{4}{3(s+1)} + \frac{1}{3(s+4)}$$

Inverse laplace of above eq^n :-

$$\mathcal{L}^{-1} C(s) = c(t) = 1 - \frac{4}{3}e^{-t} + \frac{1}{3}e^{-4t} \text{ Ans.}$$

- Ques. Consider the system shown in fig. Determine the value of K such that $\xi = 0.5$. Then Obtain the rise time t_r, peak time(t_p), M_p and t_s for unit step response.

Solution



By block diag. reduction technique,

$$\frac{C(s)}{R(s)} = \frac{16}{s^2+(0.8+16k)s+16}$$

$s^2 + (0.8 + 16k)s + 16 = 0$ ——— (1)

Comparing the eqⁿ (1) with $s^2 + 2\xi_e\omega_n s + \omega_n^2 = 0$

$\omega_n^2 = 16$

$\omega_n = 4$ rad/sec.

$2\xi_e\omega_n = 0.8 + 16k$

$\therefore k = 0.2$

$$t_r = \frac{\pi - \tan^{-1}\frac{\sqrt{1-\xi_e^2}}{\xi_e}}{\omega_n\sqrt{1-\xi_e^2}} = 0.605 \text{ sec}$$

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\xi_e^2}} = 0.906 \text{ sec}$$

$$M_p = e^{-\pi\xi_e/\sqrt{1-\xi_e^2}} \times 100 = 16.3\%$$

$$t_s = \frac{4}{\xi_e\omega_n} = \frac{4}{0.5\times 4} = 2 \text{ sec.}$$

Ques. The open loop transfer funcⁿ of a unity feedback control system is given by:-

$$G(s) = \frac{K}{s(1+sT)}$$

By what factor the amplifier gain should be multiplied so that the damping ratio is increased from 0.3 to 0.9

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{\frac{K}{s(1+sT)}}{1 + \frac{K}{s(1+sT)}\cdot 1}$$

$$\frac{C(s)}{R(s)} = \frac{K/T}{s^2 + \frac{s}{T} + \frac{K}{T}}$$

Characteristic eqⁿ is given by:-

$$s^2 + \frac{1}{T}s + \frac{K}{T} = 0$$

Compare with $s^2 + 2\xi_e \omega_n s + \omega_n^2 = 0$

$$2\xi_e \omega_n = \frac{1}{T}$$

$$\omega_n^2 = \frac{K}{T} \quad ; \quad \omega_n = \sqrt{\frac{K}{T}}$$

$$2\xi_e \sqrt{\frac{K}{T}} = \frac{1}{T}$$

$$\xi_e = \frac{1}{2\sqrt{KT}}$$

$$\xi_{e1} = 0.3 \; ; \; \xi_{e2} = 0.9$$

$$\xi_{e1} = \frac{1}{2\sqrt{K_1 T}} \, , \quad \xi_{e2} = \frac{1}{2\sqrt{K_2 T}}$$

$$\frac{\xi_{e1}}{\xi_{e2}} = \frac{1}{2\sqrt{K_1 T}} \cdot 2\sqrt{K_2 T}$$

$$\frac{\xi_{e1}}{\xi_{e2}} = \frac{\sqrt{K_2}}{\sqrt{K_1}}$$

Or $$\frac{K_2}{K_1} = \left(\frac{\xi_{e1}}{\xi_{e2}}\right)^2$$

$$\frac{K_2}{K_1} = \left(\frac{0.3}{0.9}\right)^2 = \frac{1}{9} \qquad \boxed{\therefore \; K_1 = 9K_2}$$

Hence, the gain K_1 at which $\xi_e = 0.3$ should be multiplied by $1/9$ to increase the damping ratio from 0.3 to 0.9

Ques. The open loop transfer func^n of a unity feedback system is given by $G(s) = \frac{K}{s(1+sT)}$ where K and T are constants

By what factor should the amplifier gain be reduced so that the peak overshoot of unit step response of system is reduced from 75% to 25%.

Solution

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{\frac{k}{s(1+sT)}}{1+\frac{k}{s(1+sT)}\cdot 1}$$

$$\frac{C(s)}{R(s)} = \frac{K/T}{s^2+\frac{1}{T}s+\frac{K}{T}}$$

Characteristic eqn $= s^2+\frac{1}{T}s+\frac{K}{T} = 0$ —— (1)

comparing eqn (1) with $s^2+2\xi_e\omega_n s+\omega_n^2=0$

$$2\xi_e\omega_n = \frac{1}{T} \qquad \omega_n^2 = \frac{K}{T}$$

$$\therefore \xi_e = \frac{1}{2T\omega_n} \qquad \omega_n = \sqrt{\frac{K}{T}}$$

$$\therefore \xi_e = \frac{1}{2\sqrt{KT}}$$

Given that $M_{P1} = e^{-\pi\xi_{e1}/\sqrt{1-\xi_{e1}^2}} = 0.75$

$M_{P2} = e^{-\pi\xi_{e2}/\sqrt{1-\xi_{e2}^2}} = 0.25$

$$e^{-\pi\xi_{e1}/\sqrt{1-\xi_{e1}^2}} = 0.75 \qquad e^{-\pi\xi_{e2}/\sqrt{1-\xi_{e2}^2}} = 0.25$$

$$\frac{\pi\xi_{e1}}{\sqrt{1-\xi_{e1}^2}} = 0.287 \qquad \frac{\pi\xi_{e2}}{\sqrt{1-\xi_{e2}^2}} = 1.386$$

squaring both sides and solving for ξ_{e1} and ξ_{e2}

$\xi_{e1} = 0.09118$ and $\xi_{e2} = 0.4037$

$$\frac{\xi_{e1}}{\xi_{e2}} = \sqrt{\frac{K_2}{K_1}}$$

$$\boxed{\therefore K_1 = 19.6\,K_2}$$

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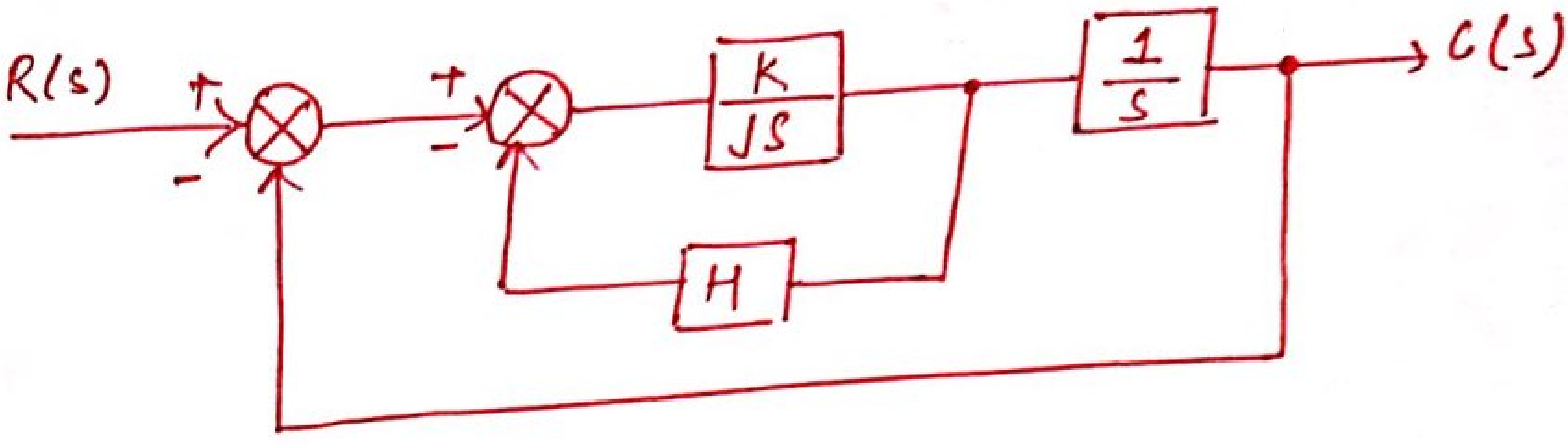
Ques. Determine the value of 'K' and 'H' of the closed loop system so that the max. overshoot in the unit step response is 25%. and peak time is 2 sec. Assume $J = 1\ kg\text{-}m^2$.



Solution :-

$$\frac{C(s)}{R(s)} = \frac{K}{JS^2 + KHS + K}$$

for $J = 1$

$$\frac{C(s)}{R(s)} = \frac{K}{S^2 + KHS + K}$$

Characteristic eqn is given by :- $S^2 + KHS + K = 0$,

compare with $S^2 + 2\xi\omega_n S + \omega_n^2 = 0$

$\therefore \omega_n^2 = K \qquad 2\xi\omega_n = KH$

$\omega_n = \sqrt{K}$

Given that $M_p = e^{-\pi\xi/\sqrt{1-\xi^2}} = 25\%$

$$e^{-\pi\xi/\sqrt{1-\xi^2}} = 0.25$$

$$\frac{-\pi\xi}{\sqrt{1-\xi^2}} = \ln(0.25)$$

$\therefore \xi = 0.404$

Also, $t_p = 2\ sec = \dfrac{\pi}{\omega_n\sqrt{1-\xi^2}}$

$$\omega_n\sqrt{1-\xi^2} = \frac{\pi}{2}$$

$$\omega_n\sqrt{1-(0.404)^2} = \frac{\pi}{2}$$

$\therefore \omega_n = 1.72\ rad/sec.$

$\therefore\ K = \omega_n^2$

$K = (1.72)^2 = 2.95$

$2\xi_0 \omega_n = KH$; $H = \frac{2\xi_0 \omega_n}{K} = \frac{2 \times 0.404 \times 1.72}{2.95}$

$\boxed{H = 0.471}$

and $\boxed{K = 2.95}$ Ans.

ERROR ANALYSIS

CLASSIFICATION OF CONTROL SYSTEM

Consider the open loop transfer func^n :-

$$G(s)H(s) = \frac{K(1+ST_1)(1+ST_2)\ldots}{S^m(1+ST_a)(1+ST_b)\ldots}$$

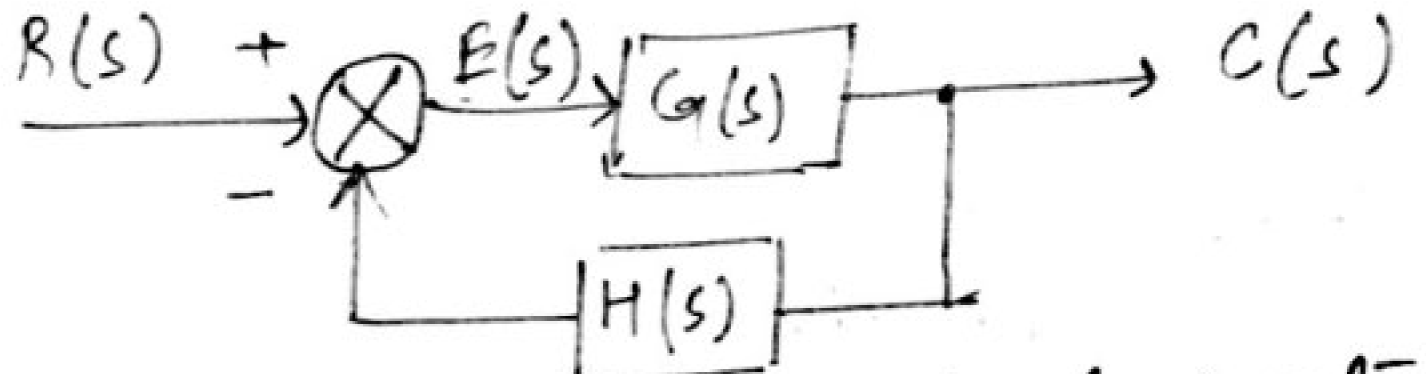
The poles are at $S = -\frac{1}{T_a}$, $S = \frac{-1}{T_b}$, ... and zeros are at $S = \frac{-1}{T_1}$, $S = \frac{-1}{T_2}$, ---

The eq^n having a term in the denominator, 'm' is the no. of poles at origion.

A system having no pole at origion of the 's' plane, is said to be type '0' system i.e. $m=0$.

A system is called type '2' system if $m=2$.

STEADY STATE ERROR



consider a closed loop control system

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)}$$

$$E(s) = \frac{R(s)}{1+G(s)H(s)}$$

Steady state error $= e_{ss} = \lim_{s\to 0} s\cdot E(s)$

$$= \lim_{s\to 0} s\cdot\frac{R(s)}{1+G(s)H(s)}$$

for unity feedback system

$$e_{ss} = \lim_{s \to 0} s \cdot \frac{R(s)}{1+G(s)}$$

STATIC ERROR COEFFICIENTS

(a) Static-Position Error Constant (K_p)

$$e_{ss} = \lim_{s \to 0} \frac{s \cdot R(s)}{1+G(s)H(s)}$$

For unit step input

$$e_{ss} = \lim_{s \to 0} \frac{\cancel{s}}{\cancel{s}} \times \frac{1}{1+G(s)H(s)}$$

$$= \lim_{s \to 0} \frac{1}{1+G(s)H(s)} = \frac{1}{1+\lim_{s \to 0} G(s)H(s)}$$

$$e_{ss} = \frac{1}{1+K_p}$$

where K_p = Static position error control $= \lim_{s \to 0} G(s)H(s)$

(b) Static Velocity Error Constant (K_v)

$$e_{ss} = \lim_{s \to 0} s \cdot R(s) \cdot \frac{1}{1+G(s)H(s)}$$

Steady state error with a unit ramp input is given by

$$e_{ss} = \lim_{s \to 0} s \cdot \frac{1}{s^2} \cdot \frac{1}{1+G(s)H(s)}$$

$$= \lim_{s \to 0} \frac{1}{s + s\,G(s)H(s)}$$

$$= \lim \frac{1}{s\,G(s)H(s)} = \frac{1}{K_v}$$

where $K_v = \lim_{s \to 0} s \cdot G(s)H(s)$ = Static velocity error coefficient (c.) Type zero system with unit parabolic input.

$$k_a = \lim_{s\to 0} s^2 G(s)H(s) = \lim_{s\to 0} \frac{s^2 k(1+sT_1)(1+sT_2)\ldots}{(1+sT_a)(1+sT_b)\ldots}$$

$$K_a = 0$$

$$e_{ss} = \frac{1}{K_a} = \infty$$

2(a) Type '1' system with unit step input $(m=1)$

$$G(s)H(s) = \frac{k(1+sT_1)(1+sT_2)\ldots}{s(1+sT_a)(1+sT_b)\ldots}$$

$$K_p = \lim_{s\to 0} G(s)H(s)$$

$$= \lim_{s\to 0} \frac{k(1+sT_1)(1+sT_2)\ldots}{s(1+sT_a)(1+sT_b)\ldots} = \infty$$

$$e_{ss} = \frac{1}{1+K_p} = 0$$

$$\boxed{e_{ss} = 0}$$

(b.) Type '1' system with unit Ramp input.

$$K_v = \lim_{s\to 0} s\cdot G(s)H(s)$$

$$= \lim_{s\to 0} \frac{\cancel{s}\cdot k(1+sT_1)(1+sT_2)\ldots}{\cancel{s}(1+sT_a)(1+sT_b)\ldots} = K$$

$$e_{ss} = \frac{1}{K_v} = \frac{1}{K}.$$

(c) Type '1' system with unit Parabolic input.

$$K_a = \lim_{s\to 0} s^2 G(s)H(s)$$

$$= \lim_{s\to 0} \frac{\cancel{s^2} k(1+sT_1)(1+sT_2)\ldots}{\cancel{s}(1+sT_a)(1+sT_b)\ldots} = 0$$

$$e_{ss} = \frac{1}{K_a} = \infty$$

(c.) Static Acceleration Error Constant (K_a)

The steady state error of the system with unit parabolic i/p is given by

$$R(s) = 1/s^3$$

$$e_{ss} = \lim_{s \to 0} s \cdot \frac{1}{s^3} \cdot \frac{1}{1+G(s)H(s)}$$

$$= \lim_{s \to 0} \frac{1}{s^2 + s^2 G(s) H(s)} = \lim_{s \to 0} \frac{1}{s^2 G(s) H(s)} = \frac{1}{K_a}$$

where $K_a = \lim_{s \to 0} s^2 G(s) H(s)$ = Static acceleration constant.

STEADY STATE ERROR FOR DIFFERENT TYPE OF SYSTEMS.

1. (a.) Type zero system with unit step i/p

$$R(s) = \frac{1}{s}$$

$$G(s)H(s) = \frac{K(1+sT_1)(1+sT_2)\ldots}{(1+sT_a)(1+sT_b)\ldots}$$

$$K_p = \lim_{s \to 0} G(s) H(s) = \lim_{s \to 0} \frac{K(1+sT_1)(1+sT_2)\ldots}{(1+sT_a)(1+sT_b)\ldots} = K$$

$$e_{ss} = \frac{1}{1+K_p}$$

$$\boxed{e_{ss} = \frac{1}{1+K}}$$

(b.) Type '0' System with Unit Ramp input

$$K_v = \lim_{s \to 0} s \cdot G(s) H(s) = \lim_{s \to 0} s \cdot \frac{K(1+sT_1)(1+sT_2)\ldots}{(1+sT_a)(1+sT_b)\ldots} = 0$$

$$e_{ss} = \frac{1}{K_v} = \frac{1}{0} = \infty$$

3. (a) Type '2' System with Unit Step Input

$$G(s)H(s) = \frac{K(1+ST_1)(1+ST_2)\ldots}{S^2(1+ST_a)(1+ST_b)\ldots}$$

$$K_p = \lim_{S\to 0} G(s)H(s)$$

$$= \lim_{S\to 0} \frac{K(1+ST_1)(1+ST_2)\ldots}{S^2(1+ST_a)(1+ST_b)\ldots} = \infty$$

$$e_{ss} = \frac{1}{1+K_p} = 0$$

(b) Type '2' System with Unit Ramp input.

$$K_v = \lim_{S\to 0} S\,G(s)H(s) = \lim_{S\to 0} \frac{S \cdot K(1+ST_1)(1+ST_2)\ldots}{S^2(1+ST_a)(1+ST_b)\ldots}$$

$$K_v = \infty$$

$$e_{ss} = \frac{1}{K_v} = 0$$

(c) Type '2' System with Unit Parabolic input

$$K_a = \lim_{S\to 0} S^2 G(s)H(s)$$

$$= \lim_{S\to 0} \frac{S^2 K(1+ST_1)(1+ST_2)\ldots}{S^2(1+ST_a)(1+ST_b)\ldots}$$

$$= K$$

$$e_{ss} = \frac{1}{K_a} = \frac{1}{K}$$

	Type '0' system	Type '1' system	Type '2' system
Unit Step Input	$\frac{1}{1+K}$	0	0
Unit Ramp Input	∞	$\frac{1}{K}$	0
Unit Parabolic Input	∞	∞	$\frac{1}{K}$

Ques 1. The open loop transfer func of unity feedback system is given by :-

$$G(s) = \frac{50}{(1+0.1s)(s+10)}$$

Determine the static error coefficients K_p, K_v and K_a.

$$K_p = \lim_{s \to 0} G(s)H(s)$$

$$= \lim_{s \to 0} \frac{50}{(1+0.1s)(s+10)} = 5$$

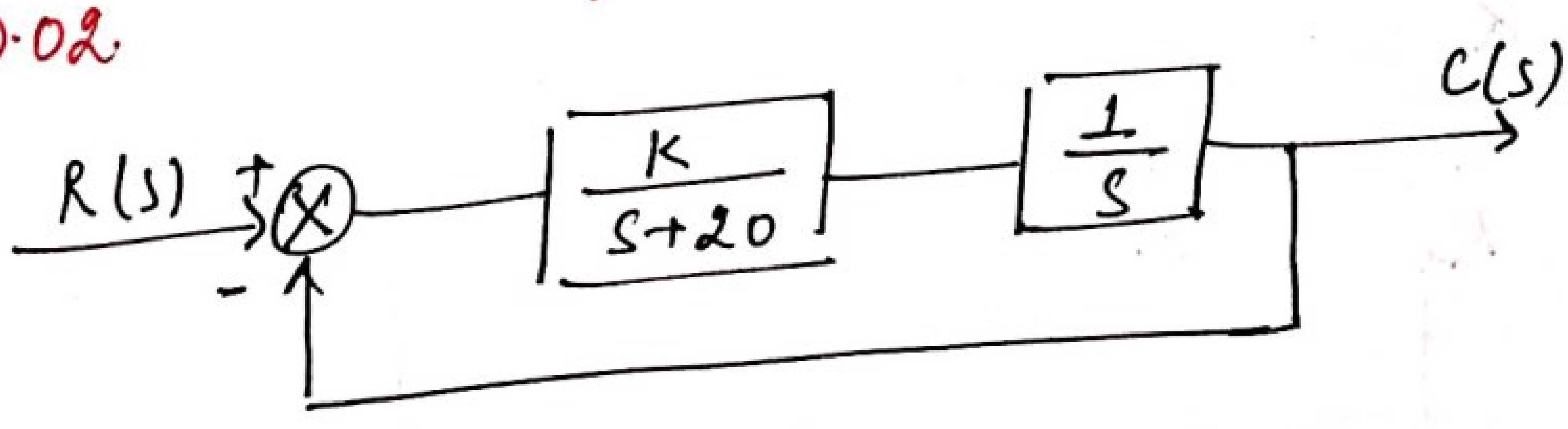
$$K_v = \lim_{s \to 0} sG(s)H(s)$$

$$= \lim_{s \to 0} \frac{s.50}{(1+0.1s)(s+10)} = 0$$

$$K_a = \lim_{s \to 0} s^2 G(s)H(s)$$

$$= \lim_{s \to 0} \frac{s^2.\,50}{(1+0.1s)(s+10)} = 0$$

Ques 2 The block diag. of an electronic pacemaker is given in fig. Determine the steady state error for unit ramp input when K = 400, Also determine the value of K for which the steady state error to a unit ramp will be 0.02.



$\cancel{\frac{C(s)}{R(s)}}$

$$K = 400$$

$$R(s) = \frac{1}{s^2}$$

$$H(s) = 1$$

$$G(s)H(s) = \frac{K}{s(s+20)}$$

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Steady state error is given by $e_{ss} = \lim_{s\to0} \frac{s\cdot R(s)}{1+G(s)H(s)}$

$$e_{ss} = \lim_{s\to0} s\cdot\frac{1}{s^2}\frac{1}{1+\frac{K}{s(s+20)}}$$

$$e_{ss} = \lim_{s\to0} \frac{s+20}{s(s+20)+400} = 0.05$$

Now $e_{ss} = 0.02$ (given)

$$e_{ss} = \lim_{s\to0} s\cdot\frac{1}{s^2}\cdot\frac{1}{1+\frac{K}{s(s+20)}}$$

$$0.02 = \lim_{s\to0} \frac{s+20}{s(s+20)+K}$$

$$0.02 = \frac{20}{K}$$

$$\boxed{\therefore K = 1000}$$ Ans.

Ques. For a unity feedback control system the forward path transfer funcn is given by :-

$$G(s) = \frac{20}{s(s+2)(s^2+2s+20)}$$

Determine the steady state error. when the inputs are :-

(i) 5 (ii) 5t (iii) $\frac{3t^2}{2}$

(i) $r(t) = 5$; $R(s) = \frac{5}{s}$

$$e_{ss} = \lim_{s\to0} s\cdot E(s) = \lim_{s\to0} s\cdot\frac{R(s)}{1+G(s)H(s)}$$

$$e_{ss} = \lim_{s\to0} s\cdot\frac{5}{s}\cdot\frac{1}{1+\frac{20}{s(s+2)(s^2+2s+20)}}$$

$$e_{ss} = \lim_{s \to 0} \frac{5s(s+2)(s^2+2s+20)}{s(s+2)(s^2+2s+20)+20}$$

$$\therefore e_{ss} = 0$$

(ii) $R(s) = \frac{5}{s^2}$

$$e_{ss} = \lim_{s \to 0} s \cdot \frac{5}{s^2} \cdot \frac{1}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$= \lim_{s \to 0} s \cdot \frac{5}{s^2} \cdot \frac{s(s+2)(s^2+2s+20)}{s(s+2)(s^2+2s+20)+20}$$

$$e_{ss} = 10$$

(iii) $R(s) = \frac{3}{s^3}$

$$e_{ss} = \lim_{s \to 0} s \cdot \frac{3}{s^3} \cdot \frac{s(s+2)(s^2+2s+20)}{s(s+2)(s^2+2s+20)+20}$$

$$e_{ss} = \infty$$

Ques. The open loop transfer funcⁿ of a unity feedback system is given by:-

$$G(s) = \frac{108}{s^2(s+4)(s^2+3s+12)}$$

Find the static error coefficients and steady state error of the system when subjected to an input given by

$$r(t) = 2 + 5t + 2t^2$$

Solution:- $K_p = \lim_{s \to 0} G(s) = \lim_{s \to 0} \frac{108}{s^2(s+4)(s^2+3s+12)} = \infty$

$$K_v = \lim_{s \to 0} sG(s) = \lim_{s \to 0} \frac{\cancel{s} \cdot 108}{s^{\cancel{2}}(s+4)(s^2+3s+12)} = \infty$$

$$K_a = \lim_{s \to 0} s^2 G(s) = \lim_{s \to 0} \frac{\cancel{s^2}\, 108}{\cancel{s^2}(s+4)(s^2+3s+12)} = \frac{108}{48}$$

$$r(t) = 2 + 5t + 2t^2$$

$$R(s) = \frac{2}{s} + \frac{5}{s^2} + \frac{4}{s^3}$$

$$e_{ss} = \frac{R_1}{1+K_p} + \frac{R_2}{K_v} + \frac{R_3}{K_a}$$

$$= \frac{2}{1+\infty} + \frac{5}{\infty} + \frac{4 \times 48}{108}$$

$$\boxed{e_{ss} = 1.77} \text{ Ans.}$$

Ques. For a unity feedback system having open-loop transfer funcⁿ as :-

$$G(s) = \frac{k(s+2)}{s^2(s^2+7s+12)}$$

Determine :- (i) type and order of system
(ii) error constants
(iii) steady state error for parbolic input.

Solution :- (i) Type 2, Order 4.

(ii) $H(s) = 1$

$$G(s)H(s) = \frac{k(s+2)}{s^2(s^2+7s+12)}$$

$$K_p = \lim_{s \to 0} G(s)H(s)$$

$$= \lim_{s \to 0} \frac{k(s+2)}{s^2(s^2+7s+12)} = \infty$$

$$K_v = \lim_{s \to 0} s\,G(s)H(s)$$

$$= \lim_{s \to 0} \frac{s\,k(s+2)}{s^2(s^2+7s+12)} = \infty$$

$$K_a = \lim_{s \to 0} s^2 G(s)H(s)$$

$$= \lim_{s \to 0} \frac{s^2 K(s+2)}{s^2(s^2+7s+12)} = \frac{K}{6}$$

(iii) For parabolic input,

$$R(s) = \frac{1}{s^2}$$

$$e_{ss} = \lim_{s \to 0} \frac{s\,R(s)}{1+G(s)H(s)}$$

$$= \lim_{s \to 0} \frac{s \cdot \frac{1}{s^2}}{1+\frac{k(s+2)}{s^2(s^2+7s+12)} \cdot 1}$$

$$= \lim_{s \to 0} \frac{s}{s^2 + \frac{K(s+2)}{s^2(s^2+7s+12)}} = 0$$

transfer funcⁿ

$$\frac{C(s)}{R(s)} = \frac{Ks+b}{s^2+as+b}.$$

Determine the open-loop transfer funcⁿ $G(s)$. Show that the steady-state error with unit ramp i/p is given by $\frac{a-k}{b}$.

For a unity feedback System, $H(s) = 1$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{G(s)}{1+G(s)}$$

$$\therefore \frac{C(s)}{R(s)} = \frac{ks+b}{s^2+as+b.} = \frac{G(s)}{1+G(s)}$$

$$(s^2+as+b)\, G(s) = (ks+b)(1+G(s))$$

$$G(s)\left[(s^2+as+b) - (ks+b)\right] = ks+b$$

$$G(s) = \frac{ks+b}{s\left[s+(a-k)\right]}$$

$$\therefore K_v = \lim_{s\to 0} s.\, G(s)H(s)$$

$$= \lim_{s\to 0} \frac{\cancel{s}(ks+b)}{\cancel{s}\left[s+(a-k)\right]}$$

$$K_v = \frac{b}{a-k}$$

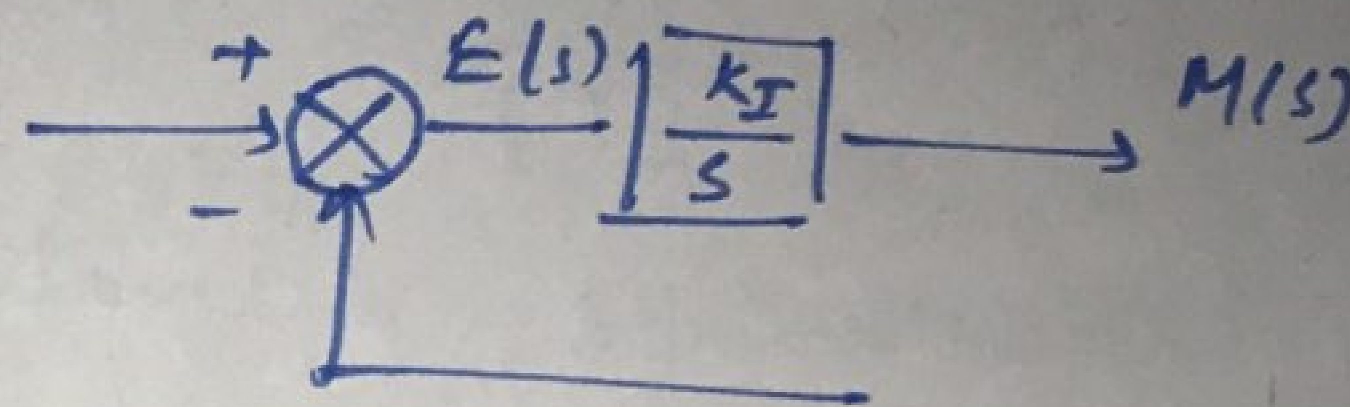
$$\therefore e_{ss}(t) = \frac{1}{K_v} = \frac{a-k}{b}.$$

Hence Proved.

INTEGRAL CONTROL ACTION

In a controller with integral control action, the output of the controller is changed at a rate which is proportional to the actuating error signal

$$\frac{d}{dt} m(t) = K_I e(t) \quad \text{——} (1)$$



where K_I is the constant

Also eq^n (1) can be written as:-

$$m(t) = K_I \int e(t) + m(0)$$

where $m(0)$ = control o/p at $t = 0$.

$$sM(s) = K_I E(s)$$

$$\frac{M(s)}{E(s)} = \frac{K_I}{s}$$

PROPORTIONAL CONTROL ACTION

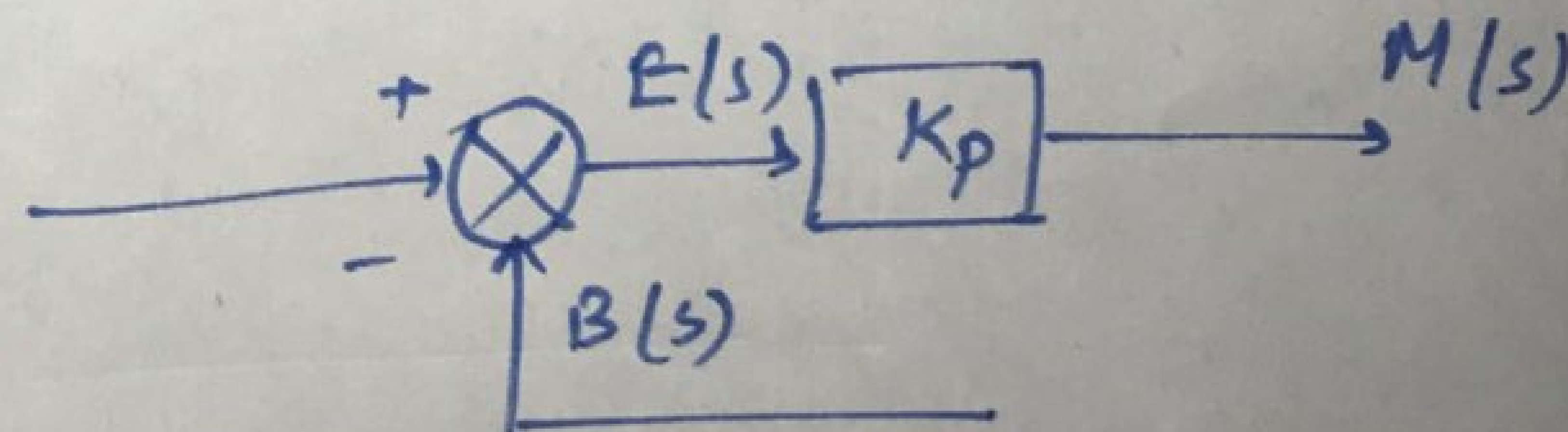
In this, there is a continous linear relationship b/w the o/p of the controller m and actuating error signal e.

Mathematically,

$$m(t) = K_p e(t) \quad \text{——} (1)$$

$$M(s) = K_p E(s)$$

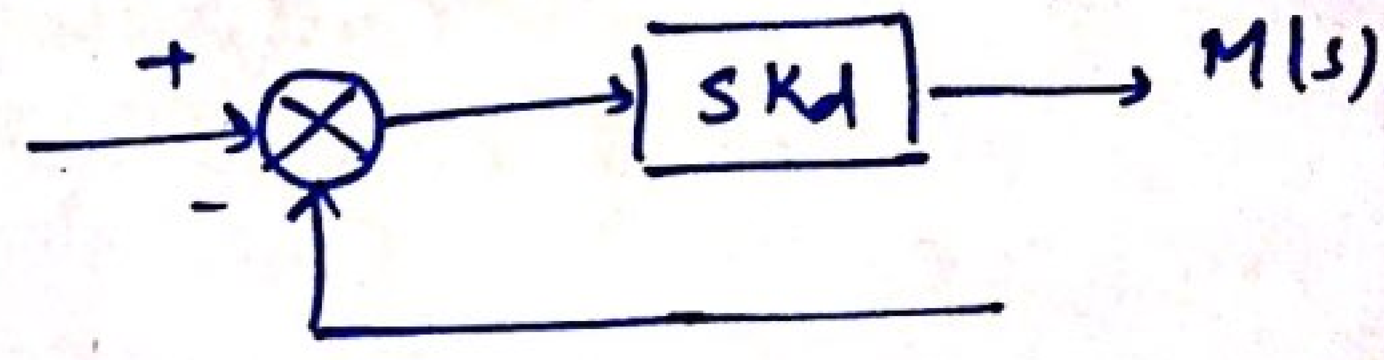
$$K_p = \frac{M(s)}{E(s)}$$



DERIVATIVE CONTROL ACTION

In this, the output of the controller depends on the rate of change of actuating error signal $e(t)$

$$m(t) = K_d \frac{d}{dt} e(t) \quad \text{—— (1)}$$



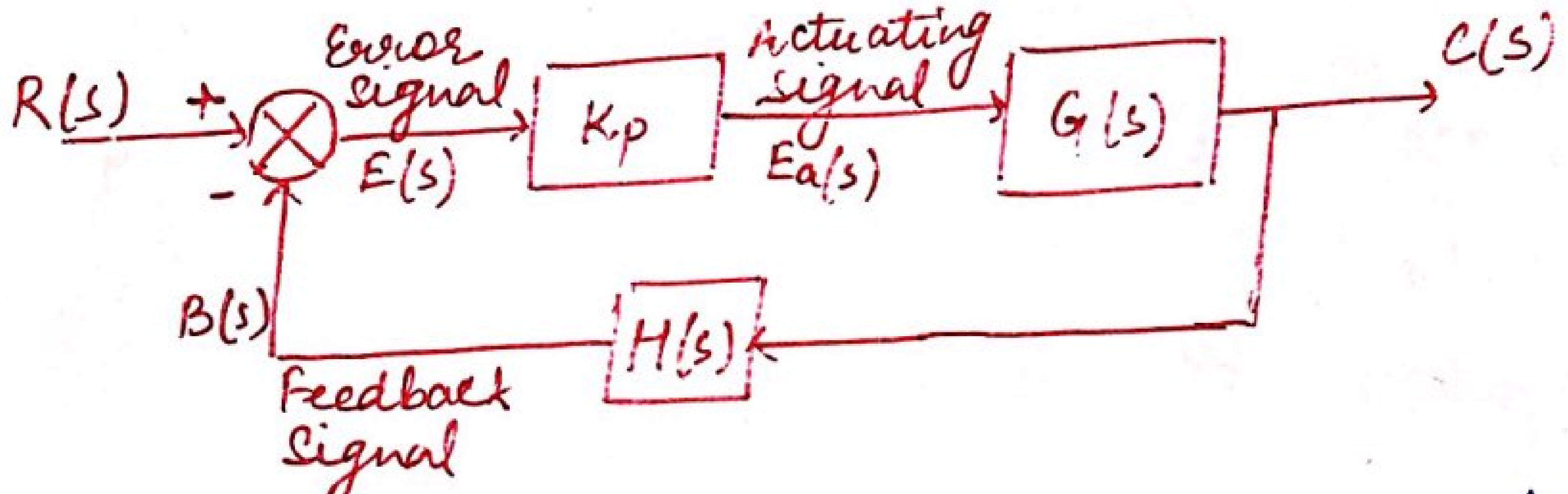
$$M(s) = K_d\, s\, E(s)$$

$$\frac{M(s)}{E(s)} = s K_d$$

where K_d = derivative gain constant

PROPORTIONAL CONTROLLER

The proportional controller is a device that produce a control signal which is proportional to the input error signal e(t). The error signal is the diff. b/w the refrence input signal and the feedback signal obtained from the output.



The above fig. shows the block diag. of a proportional control system. The actuating signal is proportional to the error signal, hence the name proportional control.

$$e_a(t) = K_p\, e(t) \quad \text{——— ①}$$

Eq^n ① represents the proportional controller in time domain. Taking laplace tranfrorm of eq^n ①

$$E_a(s) = K_p\, E(s)$$

The transform func^n of the proportional controller is

$$K(s) = \frac{E(s)}{E_a(s)} = K_p$$

The proportional controller amplifies the error signal by an amount K_p. Also, the introduction of P controller in the system increase the forward path gain by amount K_p.

PROPORTIONAL PLUS DERIVATIVE (PD) CONTROL

In proportional plus derivative controllers, the actuating signal $e_a(t)$ is proportional to the error signal $e(t)$ and also proportional to derivative of the error signal.

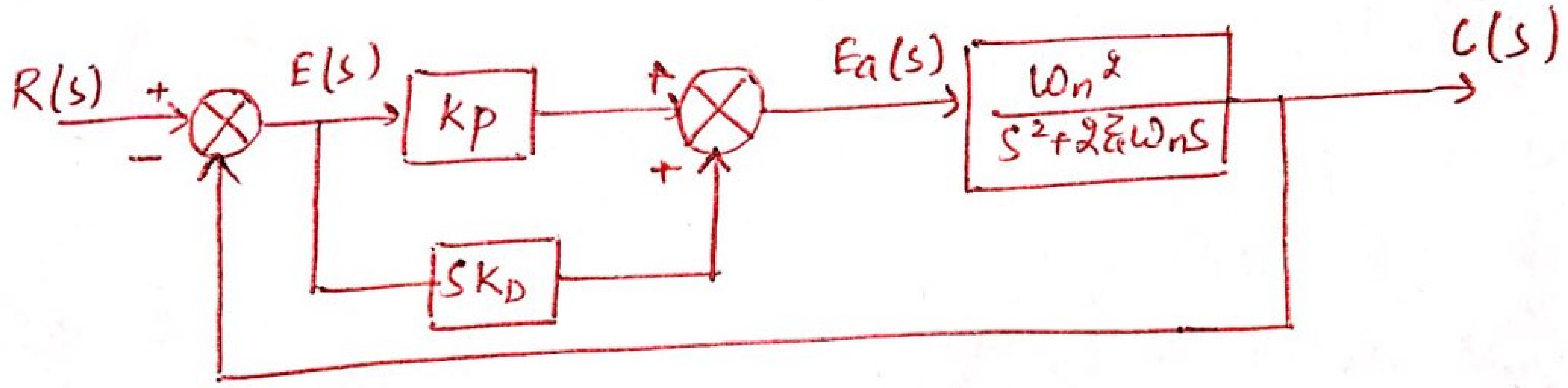
The actuating signal for proportional plus derivative control is given by:-

$$e_a(t) = K_P\, e(t) + K_D \frac{d}{dt} e(t) \quad \text{——} \quad (1)$$

Taking laplace transform on both sides of eq^n (1)

$$E_a(s) = K_P\, E(s) + K_D\, s\, E(s)$$

$$E_a(s) = (K_P + s K_D)\, E(s) \quad \text{——} \quad (2)$$



The open loop transfer func^n:-

$$G(s) = \frac{C(s)}{R(s)}$$

$$G(s) = (K_P + s K_D) \cdot \frac{\omega_n^2}{s^2 + 2\xi\omega_n s}$$

$$H(s) = 1$$

$$\therefore \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$= \frac{(K_P + s K_D)\,\omega_n^2}{s^2 + (2\xi\omega_n + K_D\,\omega_n^2)s + \omega_n^2 K_P}$$

The characteristic eq^n of the system is :-

$$s^2 + (2\xi_e\omega_n + K_D\omega_n^2)s + K_p\omega_n^2 = 0$$

Comparing the above eq^n with $s^2 + 2\xi_e\omega_n s + \omega_n^2 = 0$

$$\therefore \quad 2\xi_e\omega_n + K_D\omega_n^2 = 2\left\{\xi_e + \frac{1}{2}K_D\,\omega_n\right\}\omega_n$$

$$= 2\xi_e'\omega_n$$

$$\text{where } \boxed{\xi_e' = \left(\xi_e + \frac{1}{2}K_D\,\omega_n\right)} \quad \text{——— (3)}$$

Eq^n (3) shows that the effective damping has increased using PD control. This makes the system response slower with less overshoots increasing delay time. This PD control will not affect the steady state error of the system.

PROPORTIONAL PLUS INTEGRAL (PI) CONTROL

In PI controllers, the actuating signal consists of proportional error signal added to the integral of the error signal. The actuating signal in time domain is given by :-

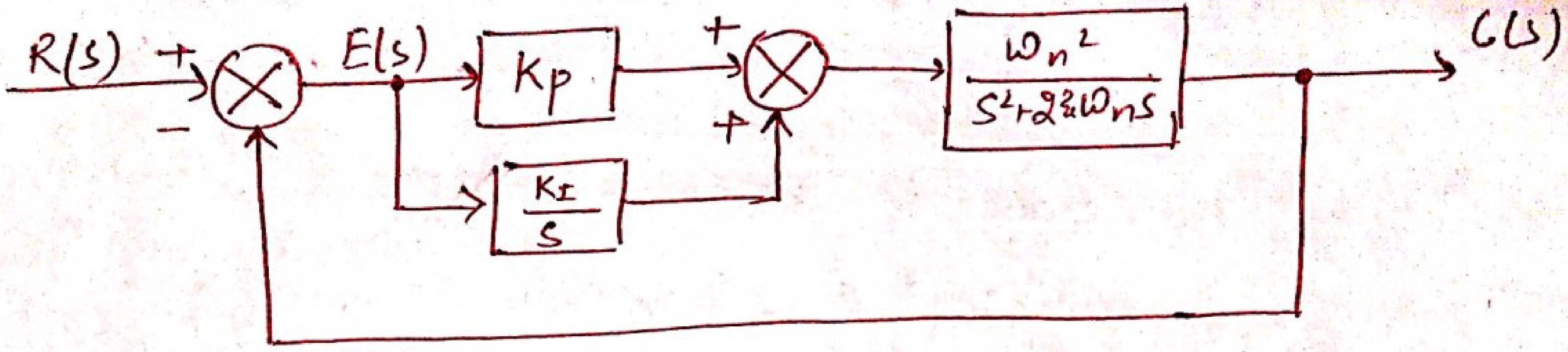
$$e_a(t) = K_p\,e(t) + K_I\int_0^t e(\tau)\,d\tau \quad \text{——— (1)}$$

Here the constants K_p and K_I are proportional and integral gains known as controller parameters.

Take L.T. of eq^n (1)

$$E_a(s) = K_p\,E(s) + K_I\,\frac{E(s)}{s}$$

$$E_a(s) = \left\{K_p + \frac{K_I}{s}\right\}E(s) \quad \text{——— (2)}$$



from above fig.,

$$G(s) = \frac{C(s)}{R(s)} = \frac{\omega_n^2\left(K_p + \frac{K_I}{s}\right)}{s^2 + 2\xi_e \omega_n s}$$

$$G(s) = \frac{(K_p s + K_I)\,\omega_n^2}{s^2 (s + 2\xi_e \omega_n)}$$

$$H(s) = 1$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$\frac{C(s)}{R(s)} = \frac{(K_p s + K_I)\,\omega_n^2}{s^3 + 2\xi_e \omega_n s^2 + K_p \omega_n^2 s + K_I \omega_n^2} \quad \text{——— (3)}$$

The characteristic eq^n is :-

$$S^2 + 2\xi_e \omega_n s^2 + k_p \omega_n^2 s + K_I \omega_n^2 = 0 \quad \text{——— (4)}$$

Eq^n (4) shows that the eq^n is of third-order system. Thus, a 2nd order system has been changed to a third-order system by adding an integral control in the system. Therefore, the effect of PI controller on the system performance is that it increases the order of the system by one, which results in reduction of the steady-state error.

PROPORTIONAL - INTEGRAL - DERIVATIVE (PID) CONTROL

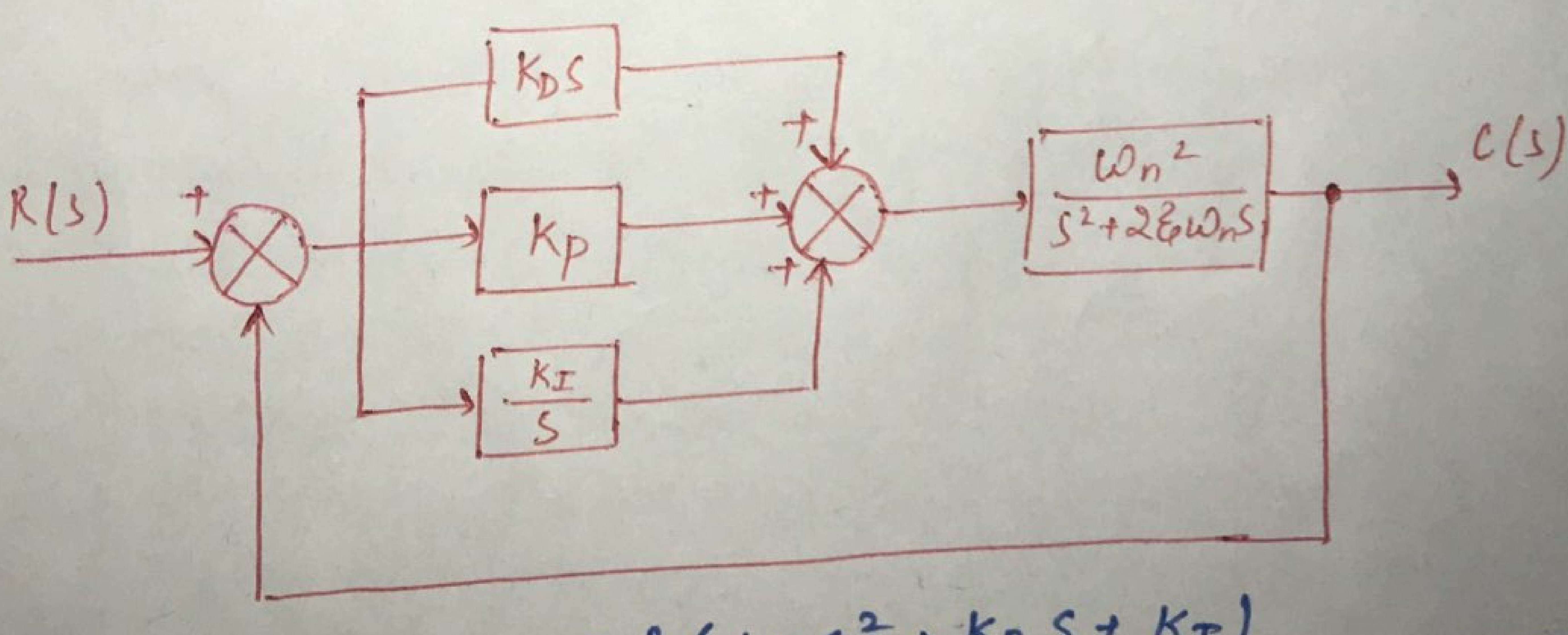
A PID controller produces an output signal consisting of three terms – one proportional to error signal, another one proportional to integral of error signal and third one proportional to derivative of error signal. The combined action of these three has the advantage of each of the three individual control actions.

The proportional controller stabilizes the gain but produces the steady state error. The integral controller reduces or eliminates steady state error. The derivative controller reduces the rate of change of error. The main advantage of PID controllers are higher stability, no offset and reduced overshoot.

$$e_a(t) = K_P e(t) + K_I \int_0^t e(\tau)\,d\tau + K_D \frac{de(t)}{dt} \quad \text{——— (1)}$$

Taking laplace transform of eq^n (1) :-

$$E_a(s) = \left\{K_P + \frac{K_I}{s} + K_D s\right\} E(s) \quad \text{——— (2)}$$



$$\frac{C(s)}{R(s)} = \frac{\omega_n^2(K_D s^2 + K_P s + K_I)}{s^3 + (2\xi\omega_n + K_D\omega_n^2)s^2 + K_P\omega_n^2 s + K_I\omega_n^2}$$

TACHOGENERATORS

Tachometers are electromechanical devices, which transforms the mechanical energy into electrical energy. In tachometers its magnetic flux is constant and induced emf is proportional to the speed of the shaft i.e. angular speed. The tachometeres are classified as d.c. tachometers and a.c tachometeres.

D.C. TACHOMETER

These contains an iron core rotor and permanent magnet. The magnetic field is provided by the permanent magnet and no external supply voltage is necessary. The input to the tachometer is the speed of the shaft and the O/P is voltage which is proportional to the angular speed of the shaft.

$$e = k\,\omega(t) \quad \text{——— (1)}$$

where, e = tachometer generator voltage

k = tachometer sensitivity

ω = angular speed of the shaft.

Laplace transform of eq^n (1) is :-

$$E(s) = k\,\omega(s)$$

$$\therefore \quad K = \frac{E(s)}{\omega(s)}$$

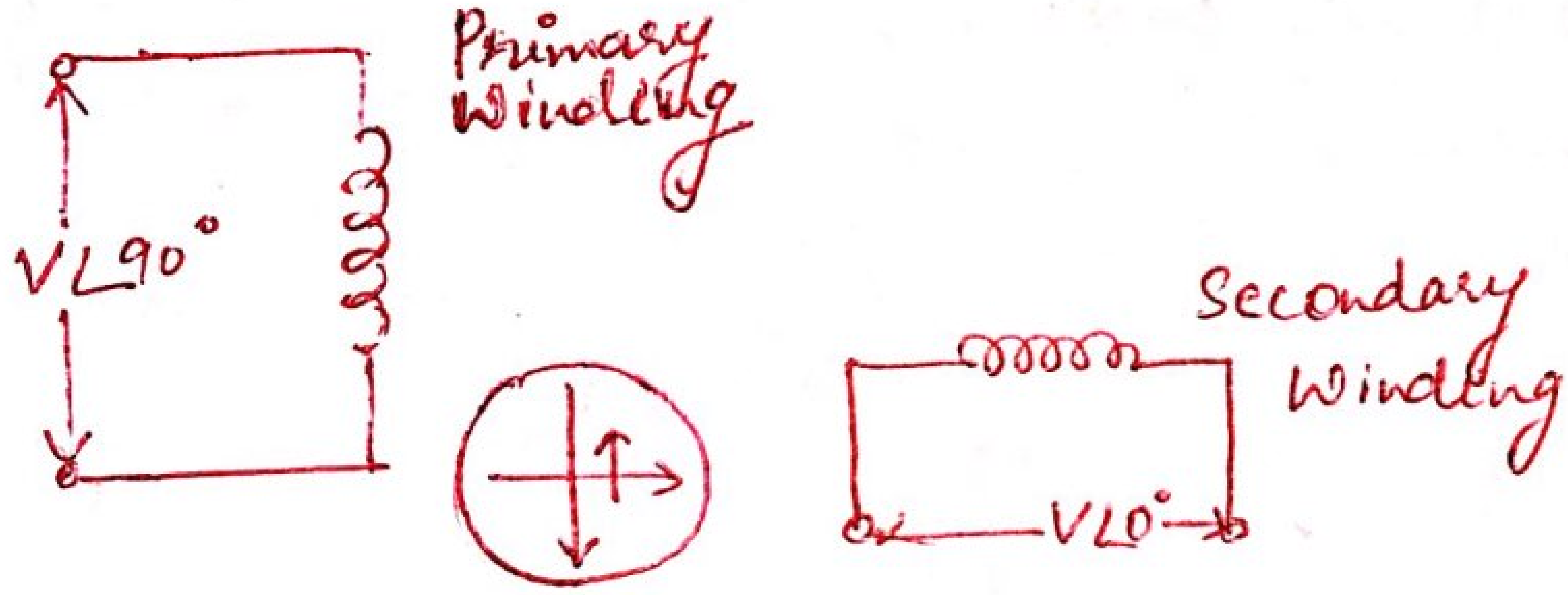
In d.c. tachometers the windings on rotor are connected to the commutator and the o/p voltage is taken across the brushes. The permanent magnet tachometers are compact & reliable but having high inertia. For reducing the voltage drop across the brushes, metal brushes with silver tips are used.

Advantages:- (i) possible to generate a very high voltage gradients in small size

(ii) At zero speed there is no residual voltage.

AC TACHOMETERS

These are similar to two-phase induction motor. In this stator windings are placed in Quadrature with each other and rotor is short circuited. The rotor of tachometer is a thin Al cup. The two stator windings are known as primary winding & secondary winding. Primary winding also known as refrence winding. The primary winding carries a voltage $V\angle 90^\circ$. This voltage is fixed in magnitude at fixed freq. known as carrier freq.



when the current in primary winding flows a pulsating field M_1 is produced along direct axis. when rotor is stationary an emf is induced in it. Since the rotor is short circuited eddy current flows in it. Due to this eddy current a pulsating flux M_2 is produced opp. to M_1. Therefore a pulsating flux ϕ_1 is produced due to vector sum of M_1 and M_2. Since the secondary winding is in Quadrature no emf will produced in it due to ϕ_1. Now if the rotor rotates it cuts the flux and an emf is induced in it due to this induced emf a current will flow in rotor and pulsating flux will appear along the coil axis of secondary winding.

CHAPTER 9

Control System Components

CHAPTER OUTLINES

■ Introduction; ■ Potentiometers; ■ Servomotors; ■ Types of Servomotors; ■ Tachogenerators (Tachometers); ■ Stepper Motors; ■ Synchros

9.1. INTRODUCTION

In control system, the devices which are used to convert the process variables in one form to another from is known as *transducers*. Transducers can also be defined as a device which transforms the energy from one form to another, for example, a thermocouple converts the heat energy into electrical voltage. In control system the following devices are used as a transducer.

1. Potentiometers
2. D.C. servomotors
3. A.C. servomotors
4. Synchros
5. Stepper motors
6. Magnetic amplifier
7. Tachogenerators
8. Gyroscope
9. Differential transformer

9.2. POTENTIOMETERS

A potentiometer is a simple device which is used for mechanical displacement either linear or angular. Thus, a potentiometer is an electromechanical transducer which converts the mechanical energy into electrical energy. The input to the device is in the form of linear mechanical displacement or rotational mechanical displacement, when the voltage is applied across the fixed terminals, the output voltage is proportional to the displacement.

Consider the Fig. 9.1 (translational potentiometer).

Let E_i = input voltage

E_0 = output voltage

x_i = displacement from zero position

x_t = total length of translational potentiometer

R = total resistance of potentiometer

Under ideal condition the output voltage E_0 is given by

$$E_0 = \frac{x_i}{x_t} E_i \qquad ...(9.1)$$

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Equation (9.1) shows a linear relationship shown in Fig. 9.2.

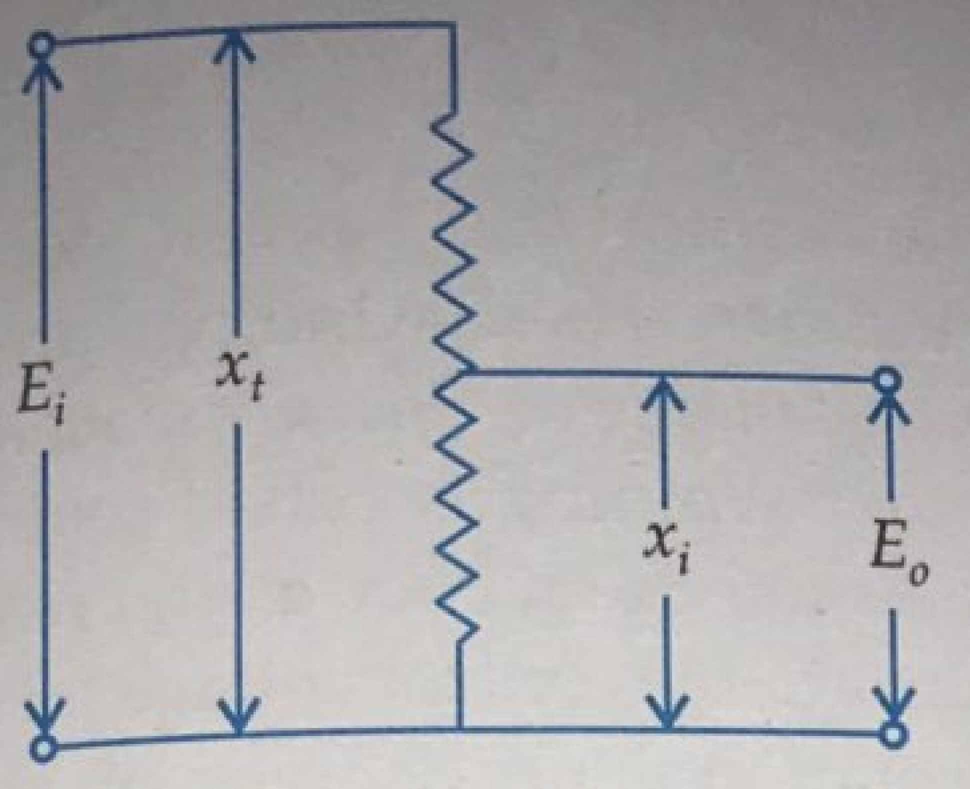


Fig. 9.1.

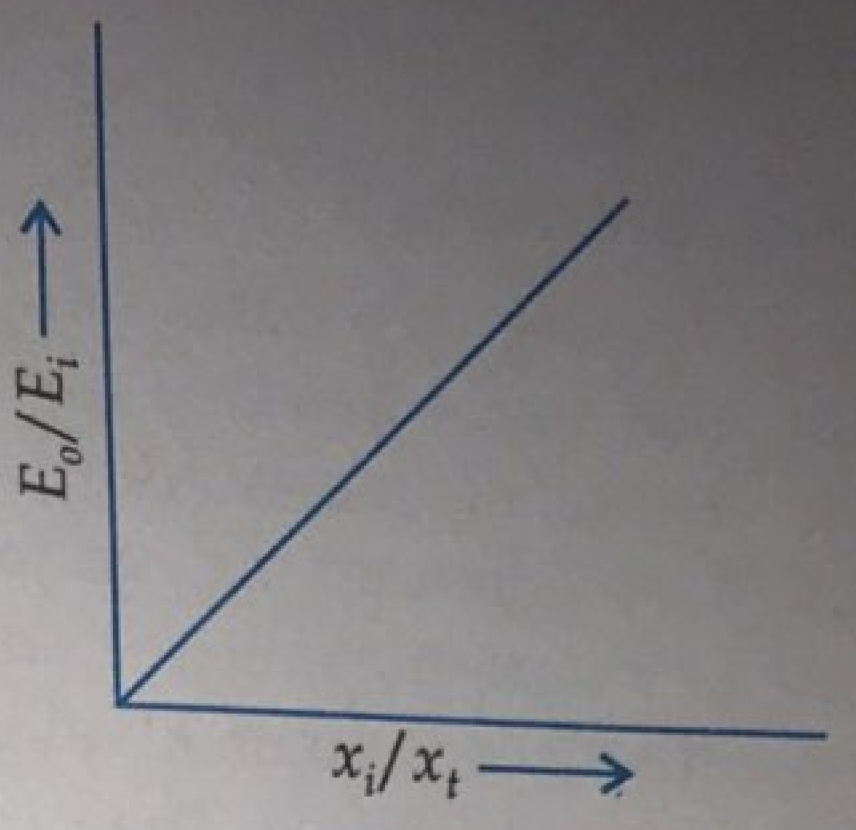


Fig. 9.2

Similarly for rotational motion, the output voltage E_0 is given by

$$E_0 = E_i \cdot \frac{\theta_i}{\theta_t} \qquad \text{...(9.2)}$$

where θ_i = input angular displacement (degree or radians)

θ_t = total travel of wiper (degree or radians)

Figure 9.3(*a*) Shows an arrangement of error sensing transducer. In this arrangement two potentiometers are connected in parallel. The output voltage taken across the variable terminals of the two potentiometer. The output voltage E_0 is given by

$$E_o = K(\theta_1 - \theta_2) \qquad \text{...(9.3)}$$

E_i is the voltage applied, θ_1 and θ_2 are the angular displacement of the wiper, K is constant and is known as sensitivity. The block diagram is shown in Fig. 9.3 (*b*).

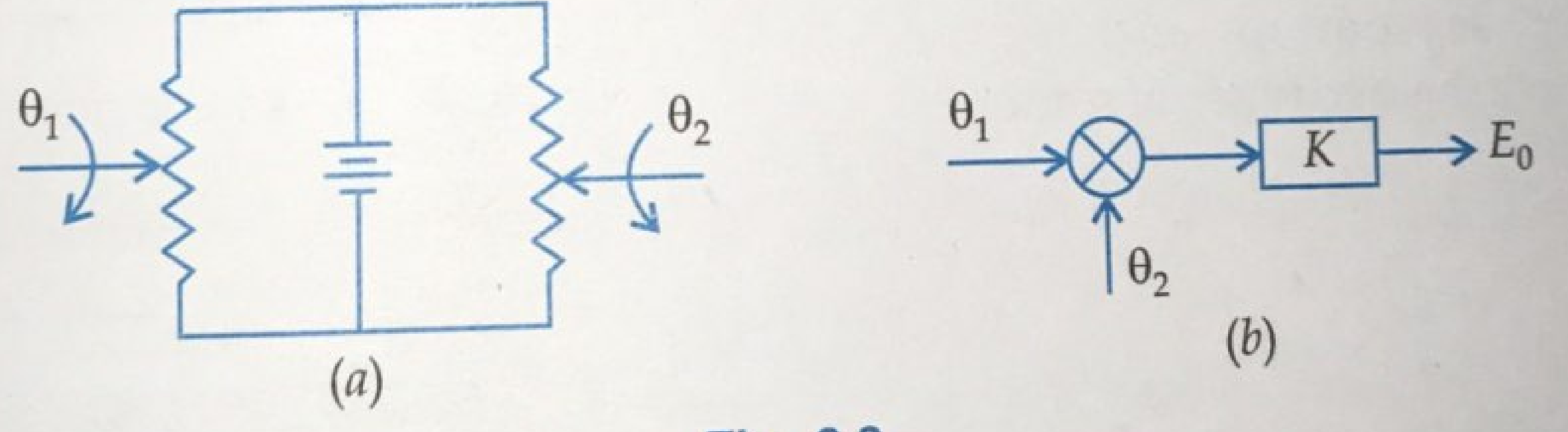


Fig. 9.3.

9.3. SERVOMOTORS

Servomotors are used in feedback control systems. Servomotors have low rotor inertia and high speed of response. The servomotors are also know as control motors. The servomotors which are used in feedback control system should have linear relationship between electrical control signal and rotor speed, torque speed characteristic should be linear, the response of the servomotor should be fast and inertia should be low.

9.4. TYPES OF SERVOMOTORS

The servomotors are classified as :

(*i*) A.C. servomotors
(*ii*) D.C. servomotors
(*iii*) Special servomotors

D.C. servomotors are further classified as armature controlled d.c. servomotors and field control d.c. servomotors.

9.4.1. A.C. Servomotors

These motors having two parts namely stator and rotor. A.C. servomotors are two phase induction motor. The stator has two distributed windings. These windings are displaced from each other by 90° electrical. One winding is called *main winding* or *reference winding*. The reference winding is excited by constant a.c. voltage. The other winding is called *control winding*. This winding is excited by variable control voltage of the same frequency as the reference winding, but having a phase displacement of 90° electrical. The variable control voltage for control winding is obtained from a servoamplifier. The direction of rotation depends upon phase relationship of voltages applied to the two windings. The direction of rotation of the rotor can be reversed by reversing the phase difference between control voltage and reference voltage.

The rotor of a.c. servomotors are of two types : (*a*) squirrel cage rotor, (*b*) drag cup type rotor. The squirrel cage rotor having large length and small diameter, so its resistance is very high. The air gap of squirrel cage is kept small. In drag cup type there are two airgaps. For the rotor a cup of non-magnetic conducting material is used. A stationary iron core is placed between the conducting cup to complete the magnetic circuit. The resistance of drag cup type is high and therefore having high starting torque. Generally aluminium is used for cup. Figure 9.4. Shows the schematic diagram of two phase a.c. servomotor and Fig. 9.5(*a*), and (*b*) shows the two types of rotor.

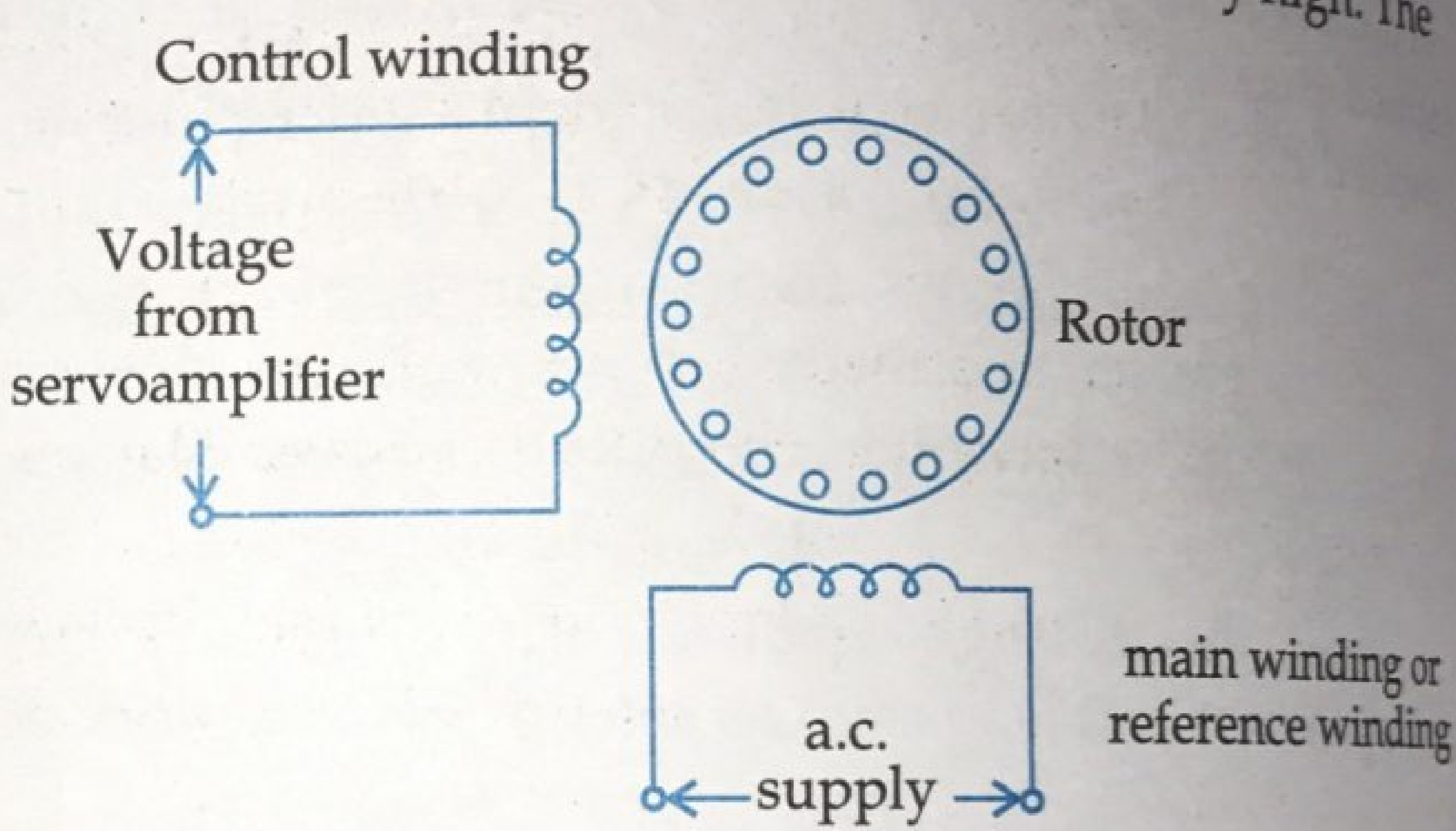


Fig. 9.4. A.C. servomotor

Fig. 9.5 (a) Squirrel cage rotor

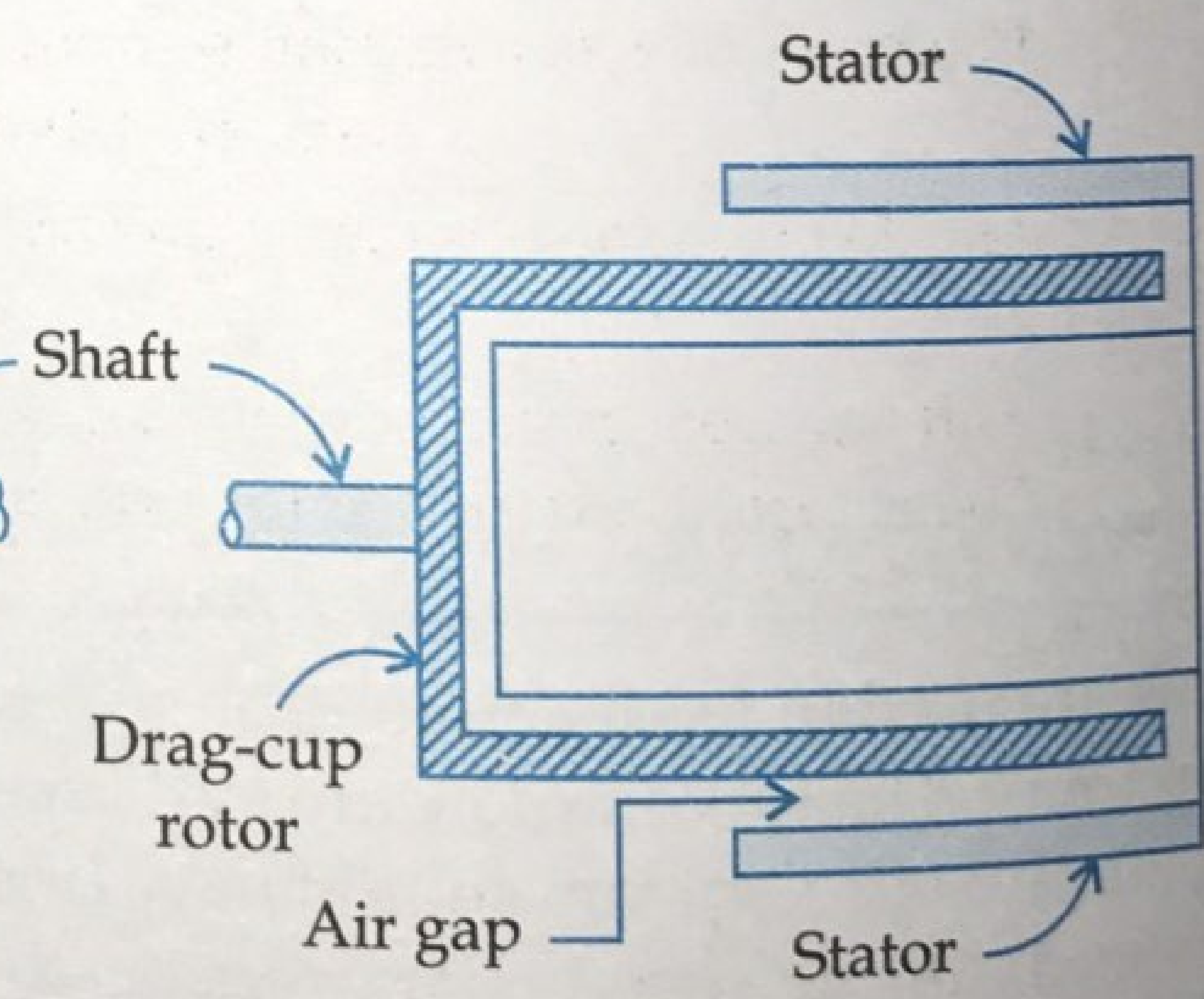


Fig. 9.5 (b) Drag cup type rotor

9.4.2. Torque-speed Characteristic

The torque speed characteristic of two phase induction motor depends upon the ratio of reactance to resistance. For high resistance and low reactance, the characteristic is linear and for large ratio of X to R it becomes non-linear as shown in Fig. 9.6(*a*). The torque-speed characteristics for various control voltages are almost linear as shown in Fig. 9.6(*b*).

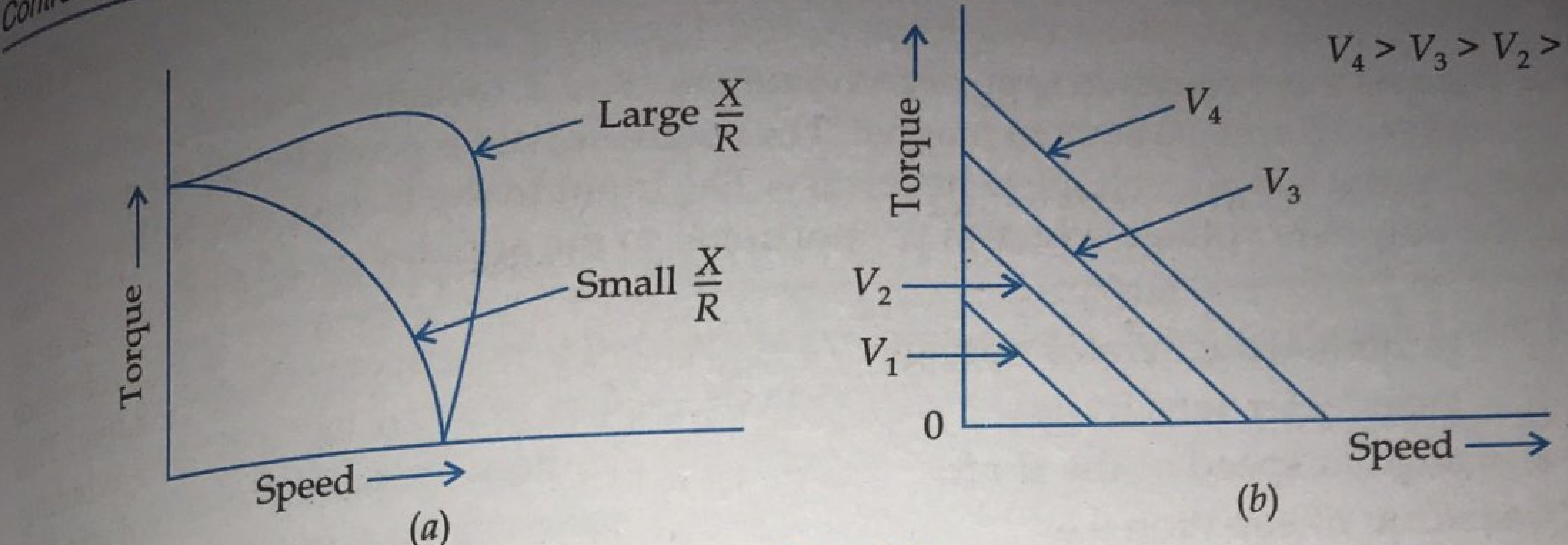
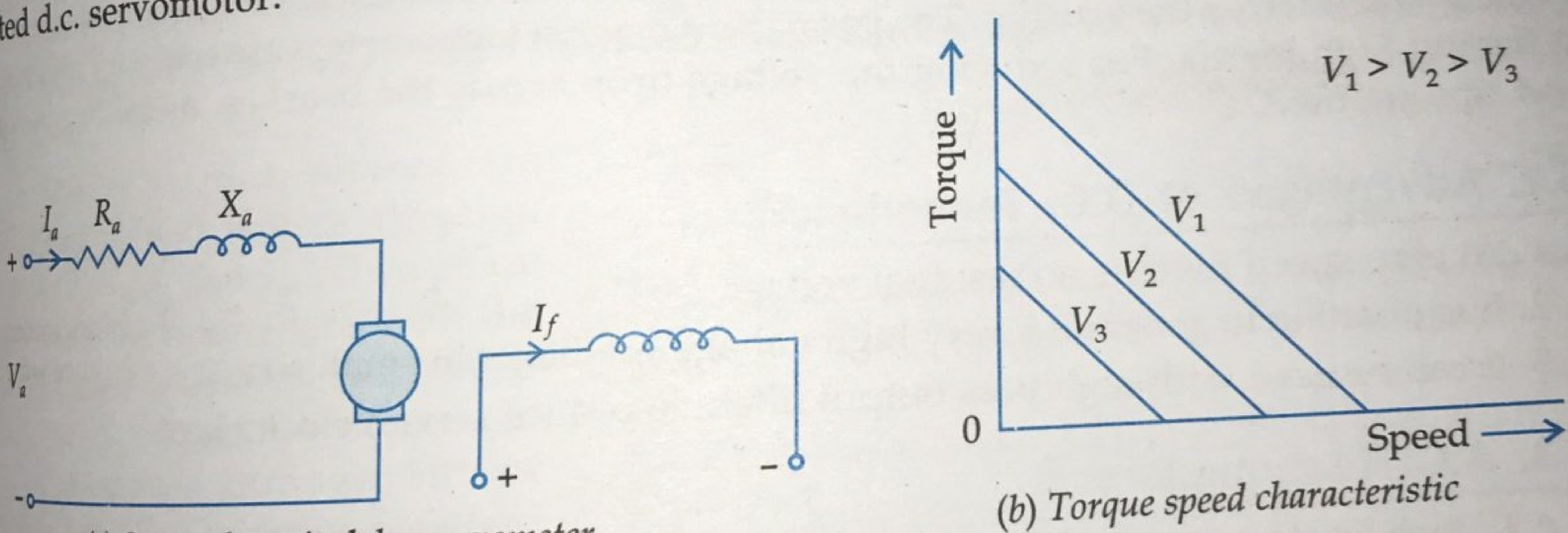


Fig. 9.6.

9.4.3. D.C. Servomotors

D.C. servomotors are separately excited or permanent magnet d.c. servomotors. The armature of d.c. servomotor has a large resistance, therefore torque speed characteristic is linear. The torque speed characteristic shows in Fig. 9.7(*b*). Fig 9.7(*a*) shows the schematic diagram of separately excited d.c. servomotor.



(*a*) *Separately excited d.c. servometer*

(*b*) *Torque speed characteristic*

Fig. 9.7.

The d.c. servomotors can be controlled from armature side or from field. In field controlled d.c. servomotors the ratio of L/R is large *i.e.*, the time constant for field circuit is large. Due to large time constant, the response is slow and therefore they are not commonly used. Transfer function of field controlled d.c. servomotor is given in Chapter 1. The speed of the motor can be controlled by adjusting the voltage applied to the armature. In armature controlled d.c. servomotor the time constant is small and hence the response is fast. The efficiency is better than the field controlled motor. The transfer function of armature controlled d.c. servomotor is derived in Chapter 1.

9.4.4. Application of Servomotors

Servomotors are widely used in radars, electromechanical actuators, computers, machin tools, tracking and guidance system, process controllers and robots.

9.6. STEPPER MOTORS

In stepper motors, the movement of the rotor is in discrete steps. A stepper motor is electromechanical device. There are three types of stepper motors:

(i) Variable reluctance motors

(ii) Permanent magnet motors

(iii) Hybrid type

9.6.1. Variable Reluctance Stepper Motors

Variable reluctance stepper motors are of two types namely, single-stack variable reluctance motor and multi stack variable reluctance motor.

9.6.1.1. Single-stack Variable Reluctance Motor

A 4-phase, 4–stator pole, 2-tooth rotor variable reluctance stepper motor is shown schematrically in Fig. 9.9.

In Fig. 9.9. four phases are connected to d.c. source through switches S_1, S_2, S_3 and S_4. When phase 1 is excited, the rotor aligns with the axis of phase 1 as shown in Fig. 9.9. If now switch S_1 off and S_2 switched on, the rotor moves through 90° in clockwise to align with the resultant air gap field which lies along the axis of phase 2. Now, phase 3 is excited and phase 2 is disconnected, the rotor aligns with the resultant airgap field which now lies along phase 3 and so on. The windings of the stator are energized in sequence 1, 2, 3, 4, 1.

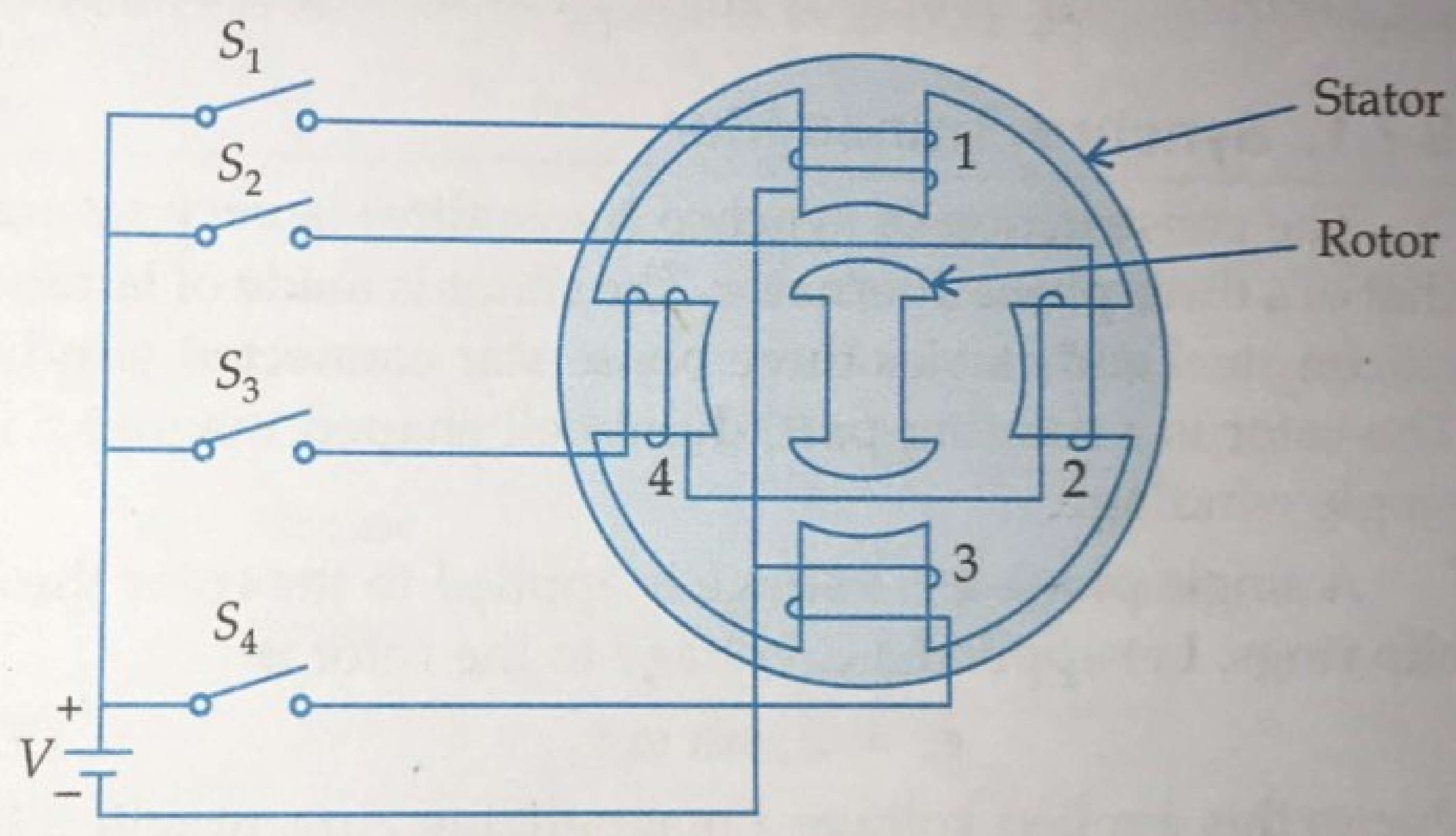


Fig. 9.9. Variable reluctance stepper motor

As the phases are energized as 1, 2, 3, 4, 1, the rotor moves through a step of 90° in clockwise direction. Rotor can be made to rotate in anticlockwise direction by reversing the switching sequence.

9.6.1.2. Multistack Variable Reluctance Stepper Motor

A multistack variable reluctance stepper motor is shown in Fig. 9.10. In this type of stepper motor, rotors have a common shaft and stator have a common frame. The stators and rotors have same number of poles with same teeth size. The stators are pulse excited while rotors are unexcited. The windings of all stator poles are excited simultaneously. Since, stators and rotors have same number poles, therefore pole pitch is also same. If T_r is the number of rotor teeth and n is the number of stacks or phases, then tooth pitch is given by $360°/T_r$ and angular displacement or step angle is given by $360°/nTr$. For exmaple 12 pole rotor, the pitch is $360/12 = 30$ and the step angle will be $360/3 \times 12 = 10°$ *i.e.*, rotor poles are displaced from each other by 10°.

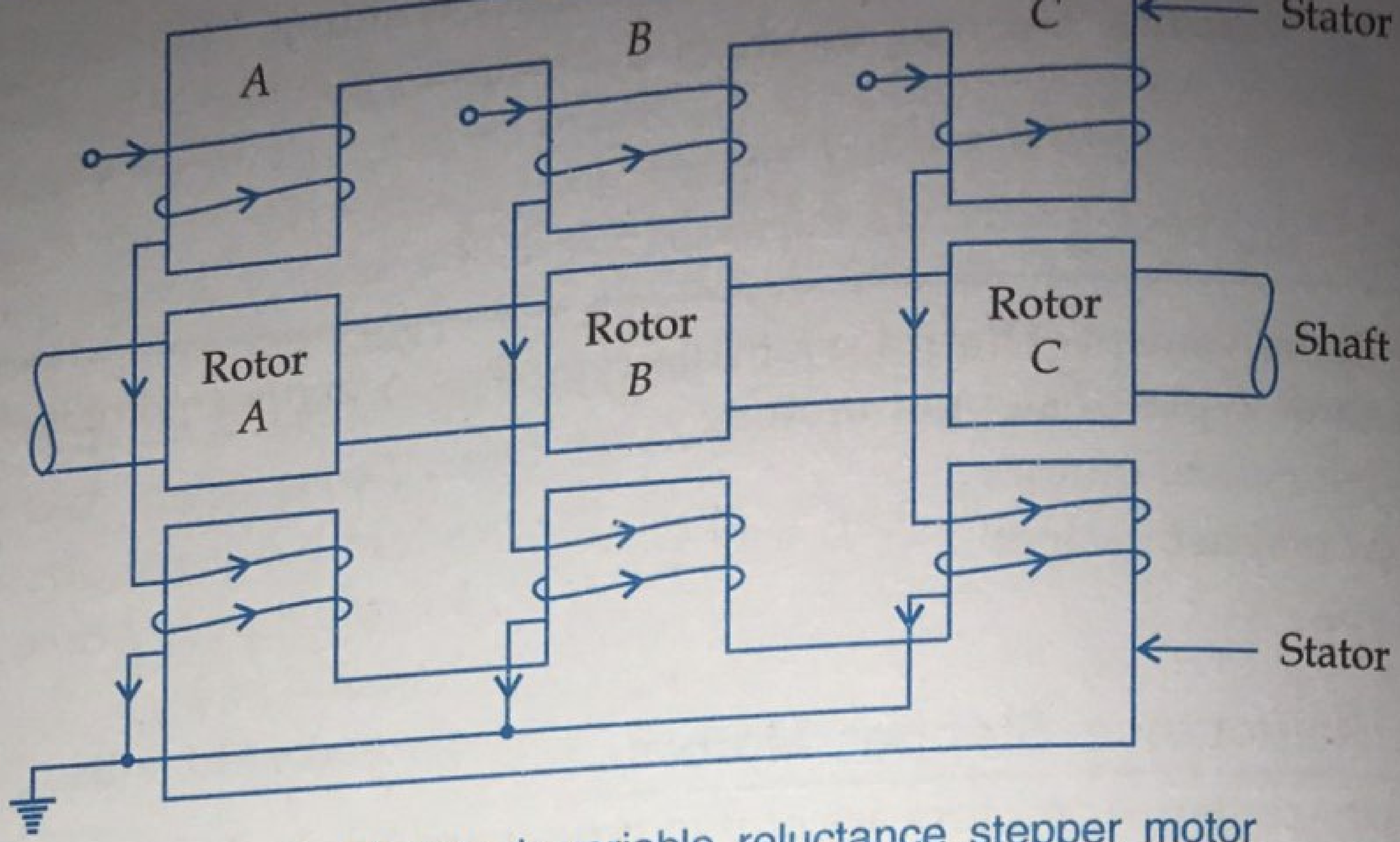


Fig. 9.10. Multistack variable reluctance stepper motor

9.7. SYNCHROS

A synchro is an electromagnetic transducer which converts the angular position of a shaft into an electric signal. Synchros are used as detectors and encoders.

9.7.1. Synchro Transmitter

The construction of synchro transmitter is very similar to that of a three phase alternator. The stator is made of laminated silicon steel and carries three phase star connected windings. The rotor is a rotating part, dumbbell shaped magnet with a single winding.

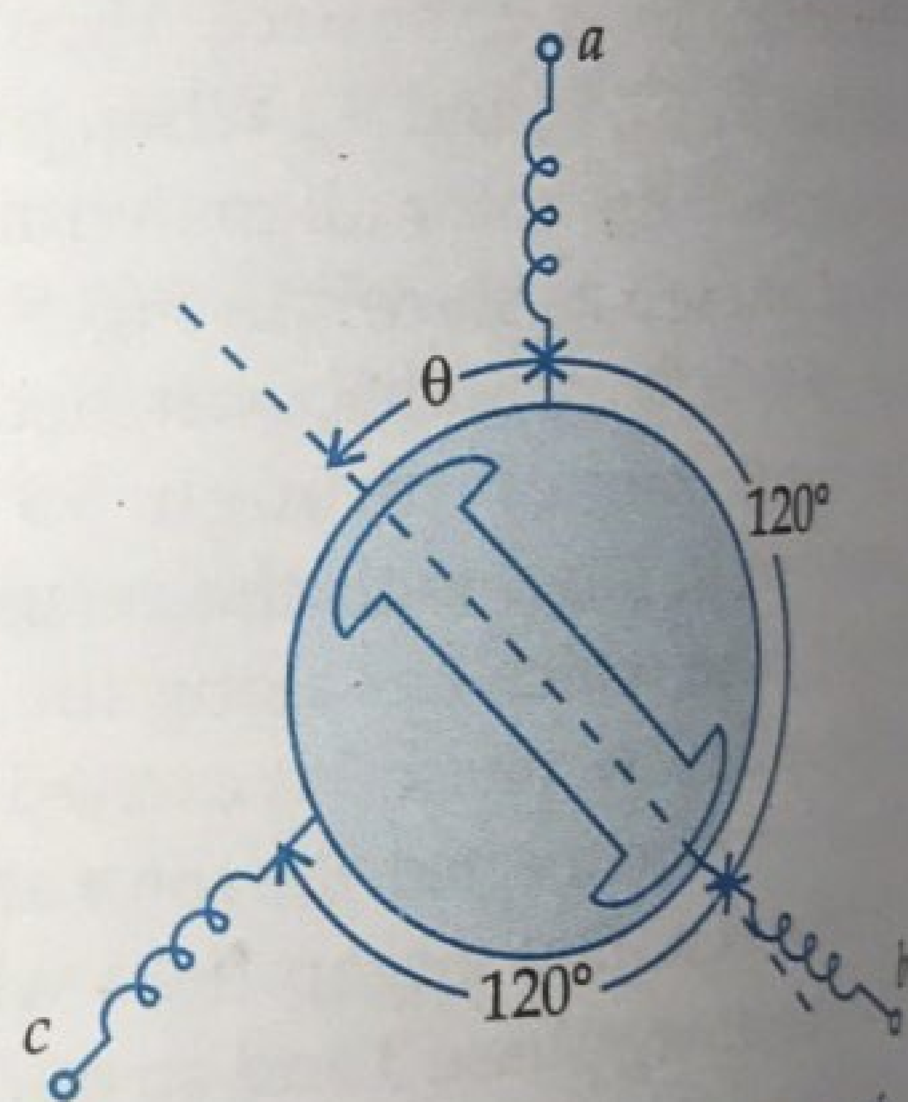


Fig. 9.11. Schematic diagram of synchro transmitter

A single phase a.c. voltage is applied to the rotor through slip rings. Let applied a.c. voltage to the rotor is

$$e_r = E_r \sin \omega_0 t \qquad ...(9.7)$$

due to this applied voltage a magnetizing current will flow in the rotor coil. This magnetizing current produces sinusoidally varying flux and distributed in the air gap. Because of transformer action voltages get induced in all stator coil which is proportional to cosine of angle between stator and rotor coil axes.

Now, consider the rotor of synchro transmitter is at an angle θ, then voltages in each stator coil with respect to neutral are

$$E_{an} = KE_r \sin \omega_0 t \cos \theta \qquad ...(9.8)$$

$$E_{bn} = KE_r \text{ son } \omega_0 t \cos (\theta + 120°) \qquad ...(9.9)$$

$$E_{cn} = KE_r \sin \omega_0 t \cos (\theta + 240°) \qquad ...(9.10)$$

Magnitudes of stator terminal voltages are

$$E_{cb} = E_{cn} - E_{bn}$$

$$= KE_r \sin \omega_0 t\, [\cos (\theta + 240°) - \cos (\theta + 120°)] = KE_r \sin \omega_0 t \left[\sqrt{3} \sin\theta\right] \qquad ...(9.11)$$

∴ $$E_{cb} = \sqrt{3}\, KE_r \sin \omega_0 t \sin \theta \qquad ...(9.12)$$

Similarly, $$E_{ac} = \sqrt{3}\, KE_r \sin \omega_0 t \sin(\theta + 120°)$$

$$E_{ba} = \sqrt{3}\, KE_r \sin\omega_0 t \sin(\theta + 240°)$$ ...(9.13)

When $\theta = 0$, the maximum induced voltage will be E_{an} and E_{cb} will be zero. This position of the rotor is defined as electrical zero of the transmitter and is used as the reference for indicating the angular position of the rotor.

Thus, the input to the synchro transmitter is the angular position of the rotor shaft and the output are the three single phase voltages which are the function of the shaft position.

9.7.2. Synchro Control Transformer

Principle of operation of synchro control transformer is same as that of synchro transmitter. Rotor of snchro control transformer is cylindrical type. Synchro control transformer is an electromechanical device. The combination of synchro transmitter and synchro control transformer is used as an error detector. The function of error detector is to convert the difference of two shaft positions into an electrical signal. The Fig. 9.12, shows schematic diagram of synchro error detector.

The output of synchro transmitter is connected to the stator winding of the synchro control transformer. Therefore the same current will flow in the stator windings of synchro control transformer but in opposite direction. The voltage across the rotor terminals of control transformer is

$$e(t) = K_1 V_r \cos\phi \sin\omega_0 t$$ ...(9.14)

where ϕ = angular displacement between the two rotors. When the two rotors are at an angle 90°, the voltage induced in control trnsformer is zero. This position is known as electrical zero position control transformer.

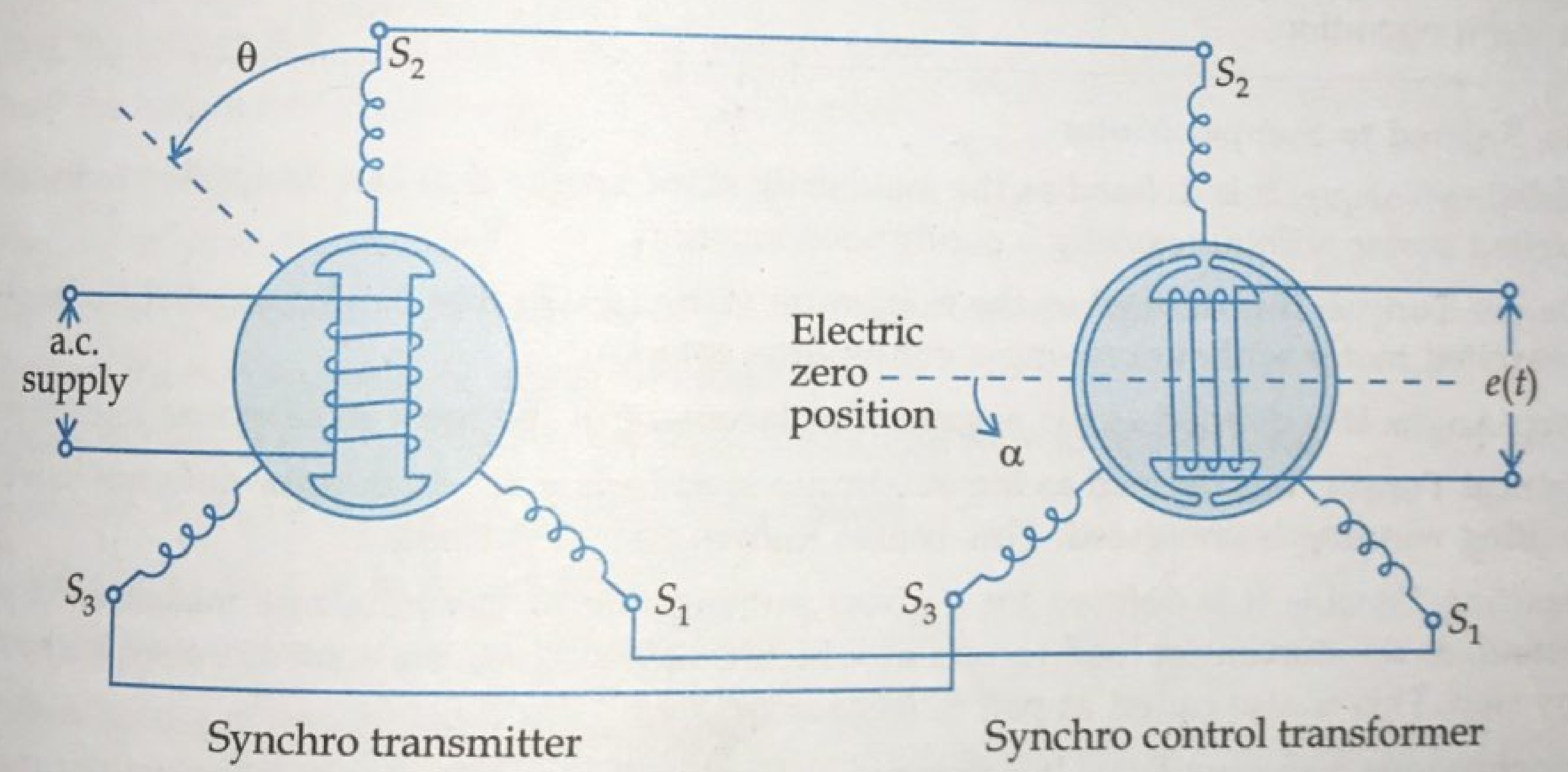


Fig. 9.12. Synchro error detector

Let the transmitter rotate through an angle 'θ' in the direction indicated and let control transformer rotor rotates in the same direction through an angle 'α'. Then

$$\phi = (90° - \theta + \alpha)$$ ...(9.15)

Put the value of ϕ in equation 9.14, we get

$$e(t) = K_1 V_r \sin(\theta - \alpha) \sin\omega_0 t$$ ...(9.16)

From equation (9.16) it is clear that when two rotor shafts are not in alignment, the rotor voltage of control transformer is approximately a sine function of the difference between the two shaft angles.

For small angular displacement between two rotor position

$$e(t) = K_1 V_r (\theta - \alpha) \sin\omega_0 t$$ ...(9.17)